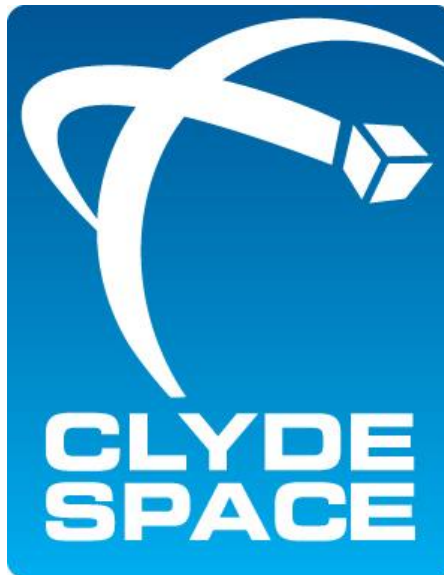




User Manual: 3rd Generation EPS Range - No Inhibits

Document No.: USM-1335

Issue: F

Date: 25 Nov 2019

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SOLUTIONS FOR A NEW AGE IN SPACE

Document Control

Issue	Date	Section	Description of Change	Reason for Change
A	05 Apr 2016	All	Information from USM-25-01309, USM-25-01311 RevF, USM-01-01317 RevE, USM-25-02452 RevC, USM-01-02453 RevB combined to create family user manual	
B	13 Jun 2017	2 3 9	Updating the table in section Update figure 3.1 Adding section for 5V USB Charge	DCR039
C	20 Jun 2017	9.8 11.4	Updated the trip current for switches 3 and 4 in Table 9-1. Updated the telemetry equations (ISW3, ISW4, VBCR1, VBCR2, VBCR4, VBCR5, VBCR6, VBCR7, VBCR8 and VBCR9) in tables 11-7, 11-9 & 11-10)	DCR044
D	02 Aug 2017	11.2 11.3	Get Version+Revision is now split into two separate commands for devices running firmware version 18-02012 Rev E + 18-2013 Rev C and above	DCR73
E	02 Nov 2018	9.8 11.4	Corrected unit for latch-off time in table 9-1 from h to ms. Corrected telemetry equations in tables 11-8, 11-9, and 11-11.	DCR218
F	25 Nov 2019	11.4	Updating table 11-8 for IPCM12V, IPCMBATV, IPCM3V3 and IPCM5V due to INA load resistor change	

Revision Control

Product	Part Number	Notes
3 rd Generation EPS (1UB) No Inhibits	25-02451	N/A
3 rd Generation EPS (3UA) No Inhibits	25-02452	H-K
3 rd Generation EPS (XUA) No Inhibits	01-02453	G

Acronyms and Abbreviations

1U	1 Unit (Cubesat standard size)
2s1p	Battery configuration – 2 cells in series, 1 string in parallel
2s2p	Battery configuration – 2 cells in series, 2 string in parallel
2s3p	Battery configuration – 2 cells in series, 3 string in parallel
3U	3 Unit (Cubesat standard size)
ADC	Analogue to Digital Converter

Ah	Ampere Hour
AIT	Assembly, Integration and Testing
AMUX	Analogue Multiplexer
BCR	Battery Charge Regulator
DoD	Depth of Discharge
EoC	End of Charge
EPS	Electrical Power System
ESD	Electro Static Discharge
FlexU/XU	FlexIble Unit (suitable for various satellite configurations)
Isc	Short Circuit Current
Kbits ⁻¹	Kilobits per second
MPPT	Maximum Power Point Tracker
PCM	Power Conditioning Module
PDM	Power Distribution Module
rh	Relative Humidity
TLM	Telemetry
USB	Universal Serial Bus
Voc	Open Circuit Voltage
Wh	Watt Hour

Related Documents

No.	Document Name	Doc Ref.
RD-1	3rd Generation CubeSat Battery Family Rev B User Manual	USM-1192
RD-2	CubeSat Design Specification	CubeSat Design Specification Rev. 12
RD-3	NASA General Environmental Verification Standard	GSFC-STD-7000 April 2005
RD-4	CubeSat Kit Manual	UM-3
RD-5	Solar Panel User Document	TBC
RD-6	Power System Design and Performance on the World's Most Advanced In-Orbit Nanosatellite	As named

 Warning 	Risk
Ensure headers H1 and H2 are correctly aligned before mating boards	If misaligned, battery positive can short to ground, causing failure of the battery and EPS
Ensure switching configuration is implemented correctly before applying power to EPS	If power is applied with incorrect switch configuration, the output of the BCR can be blown, causing failure of the EPS
Observe ESD precautions at all times	The EPS is a static sensitive system. Failure to observe ESD precautions can result in failure of the EPS.
Ensure not to exceed the maximum stated limits	Exceeding any of the stated maximum limits can result in failure of the EPS
Ensure batteries are fully isolated during storage	If not fully isolated (by switch configuration or separation) the battery may over-discharge, resulting in failure of the battery
No connection should be made to H2.35-36 or H2.41-44	These pins are used to connect the battery to the EPS. Any connections to the unregulated battery bus should be made to pins H2.45-46
H1 and H2 pins should not be shorted at any time	These headers have exposed live pins which should not be shorted at any time. Particular care should be taken regarding the surfaces these are placed on.

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1. INTRODUCTION

This document provides information on the features, operation, handling and storage of the 25-02451, 25-02452 and 01-02453 EPS products, designed to integrate with a suitable battery and solar arrays to form a complete power system for use on a CubeSat.

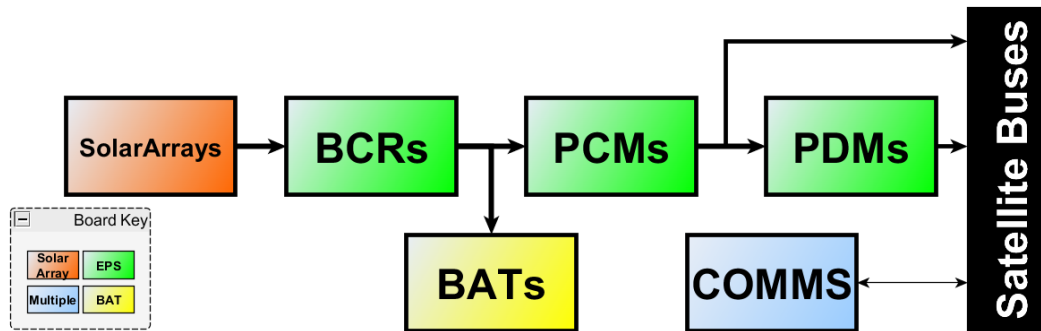


Figure 1-1 System Diagram

1.1 Additional Information Available Online

Additional information on CubeSats and Clyde Space Systems can be found at www.clyde.space. You will need to log in to our website to access certain documents.

1.2 Continuous Improvement

At Clyde Space we are continuously improving our processes and products. We aim to provide full visibility of the changes and updates that we make, and information of these changes can be found by logging in to our website: www.clyde.space

1.3 Document Revisions

In addition to hardware and software updates, we also update make regular updates to our documentation and online information.

2. OVERVIEW

The Clyde Space 3rd Generation (3G) EPS range is the latest incarnation of Clyde Space's highly successful CubeSat power system range, which has powered almost half of all CubeSat missions to date. Our 3G range is our most capable and compact CubeSat EPS range to date, providing robust, high-performance mission capability. The main features include:

- 3.3V, 5V, and 12V regulated power buses
- Unregulated battery bus
- 10 Latching Current Limit (LCL) power distribution modules
- Maximum Power-Point Tracking (MPPT) Battery Charge Regulators (BCRs)
- Over-current, over- and under-voltage protection
- Watchdog timer

As a result of the new features added in the 3rd Generation there is a requirement to alter the interfaces to the main CubeSat Kit header. Further detail on the new features and interfaces can be found in this user manual.

Clyde Space is a world-leading provider of spacecraft power systems, from CubeSats through to Small GEO satellites. Since establishment in 2006 Clyde Space has provided thousands of spacecraft subsystems, and has grown to be a leading provider of nanosatellite spacecraft platforms as well. Our heritage and experience mean you can be sure a Clyde Space component comes with performance and quality assured.

2.1 Applicable Products

This user manual describes three variants of the 3rd Generation No Inhibits EPS. The differences between these variants are summarised below

	3G 1U EPS (25-02451)	3G 3U EPS (25-02452)	3G FlexU EPS (01-02453)
Intended Use	1U CubeSats	2U and 3U CubeSats with body panels	3U CubeSats with deployable panels, and larger nanosatellites
Number of SEPIC BCRs	4 BCRs	1 BCR	1 BCR
Number of Buck BCRs	None	2 BCRs	8 BCRs

3. MAXIMUM RATINGS

Stresses beyond those listed under maximum ratings may cause permanent damage to the EPS. Operation of the EPS at conditions beyond those indicated is not recommended. Exposure to absolute maximum ratings for extended periods may affect EPS reliability. De-rating of power critical components is in accordance with ECSS guidelines.

OVER OPERATING TEMPERATURE RANGE (UNLESS OTHERWISE STATED)				
			Value	Unit
Input Voltage ⁽²⁾	Buck BCRs		30	V
	SEPIC BCRs		9.5	V
	Battery		8.3	V
			Value	Unit
Input Current	Buck BCRs		Refer to Section 3.1	
	SEPIC BCRs			
			Value	Unit
Operating Temperature			-40 to 85	°C
Storage Temperature			-50 to 100	°C
Vacuum			10 ⁻⁵	torr
Radiation Tolerance			10	kRad
Vibration			To [RD-3]	

Table 3-1 Max Ratings of the EPS products

3.1 BCR Safe Operating Area

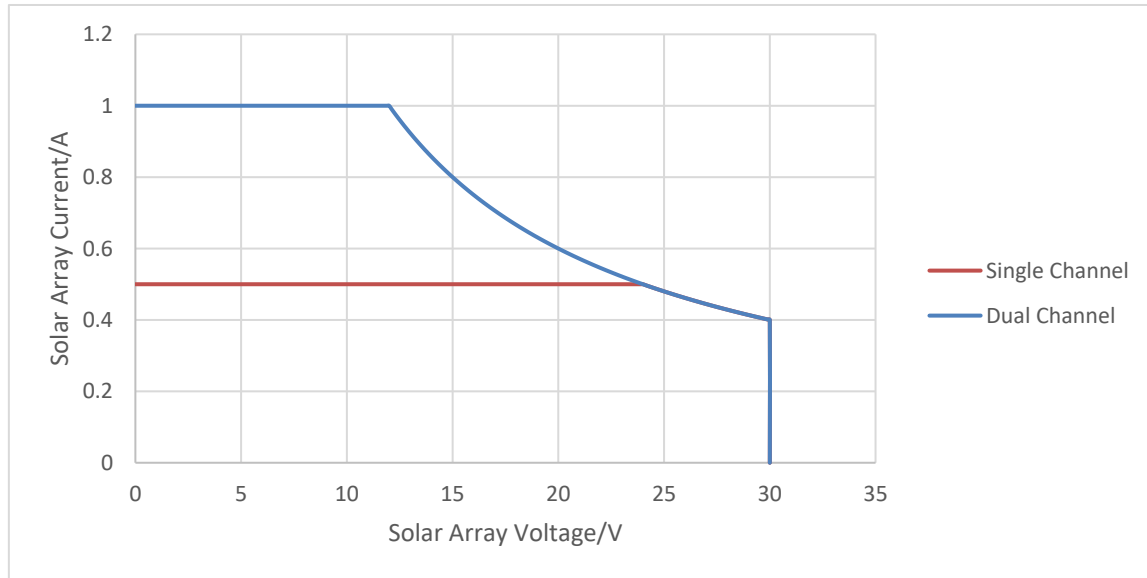


Figure 3-1 Safe Operating Range for Buck BCRs

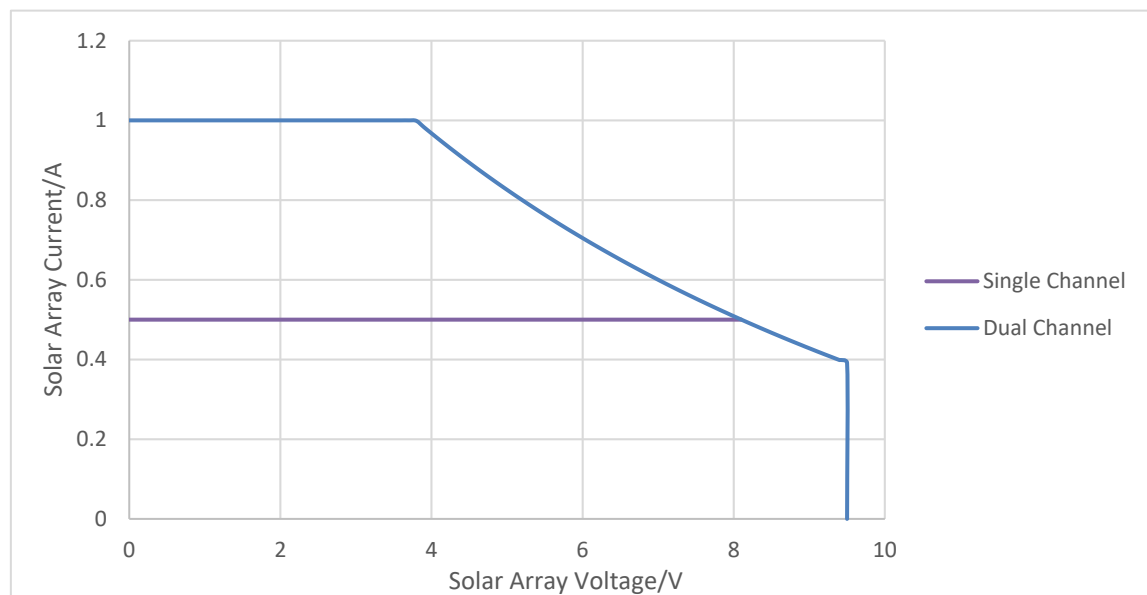


Figure 3-2 Safe Operating Range for SEPIC BCRs at $T_{BRD}=60^{\circ}\text{C}$

The safe operating ranges of the BCRs are shown above. Single Channel refers to the maximum current which can be applied to a single pin (e.g. SA1.1). Dual Channel refers to the limit on the sum of the currents which can be applied to two pins connecting to the same BCR (e.g. SA1A.1 and SA1B.1 or SA5.1 and SA5.5). For BCR allocations, see Section 9.2. It is important to ensure that the limits of the BCRs given above are not exceeded either at open circuit or at maximum power point.

4. ELECTRICAL CHARACTERISTICS

Description	Conditions	Min	Typical	Max	Unit
Buck BCRs					
Input Voltage		7.4 ¹	--	30	V
End of Charge Voltage		8.165	8.265	8.365	V
Switching Frequency		150	175	200	kHz
Efficiency	@16.5V input, Full Load	85%	90%	92%	
SEPIC BCRs					
Input Voltage		3.0	--	9.5	V
End of Charge Voltage		8.165	8.265	8.365	V
Operating Frequency		145	170	200	kHz
Efficiency	@6V input, Full Load	77%	79%	80%	
Unregulated Battery Bus					
Output Voltage		--	Battery Voltage	--	
LCL Trip Point		4.55	4.7	4.85	A
Efficiency	@8.26V input, Full Load	98.5%	99%	99.5%	
5V Bus					
Output Voltage		4.95	5	5.05	V
LCL Trip Point		4.4	4.5	4.6	A
Operating Frequency		400	480	560	kHz
Efficiency	@5V, Full Load	--	93%	--	
3.3V Bus					
Output Voltage		3.267	3.3	3.333	V
LCL Trip Point		4.4	4.5	4.6	A
Operating Frequency		400	480	560	kHz
Efficiency	@3.3V, Full Load	--	90%	--	
12V Bus					
Output Voltage		11.88	12	12.12	V
LCL Trip Point		1.4	1.5	1.6	A
Operating Frequency		670	800	930	kHz
Efficiency	@3.3V input, Full Load	--	92%	--	
Communications					
Protocol		--	I ² C	--	
Transmission speed		--	100	--	kbits ⁻¹
Bus voltage		3.26V	3.3V	3.33V	
Node address		--	0x2B	--	
Address scheme		--	7 bit	--	
Node operating frequency		--	27MHz	--	
Quiescent Operation					
Power Draw	Flight Configuration of Activation Switches	25-02451:	0.2		W
		25-02452:	0.2		
		01-02453:	0.4		

Table 4-1 Performance Characteristics of the EPS

¹If V_{MPP} is below 9.4V, the converter will deviate from the maximum power point during taper charge and will charge less efficiently

5. HANDLING AND STORAGE

The EPS requires specific guidelines to be observed for handling, transportation and storage. These are stated below. Failure to follow these guidelines may result in damage to the units or degradation in performance.

5.1 Electro Static Discharge (ESD) Protection

The EPS incorporates static sensitive devices and care should be taken during handling. Do not touch the EPS without proper electrostatic protection in place. All handling of the system should be done in a static dissipative environment.

5.2 General Handling

The EPS is robust and designed to withstand flight conditions. However, care must be taken when handling the device. Do not drop the device as this can damage the EPS. There are live connections between the battery systems and the EPS on the CubeSat Kit headers. All metal objects (including probes) should be kept clear of these headers.

Gloves should be worn when handling all flight hardware.

Flight hardware will be delivered conformally coated, and should only be removed from packaging in a class 100000 (or better) clean room environment.

5.3 Shipping and Storage

The devices are shipped in anti-static packaging, enclosed in a hard protective case. This case should be used for storage. All hardware should be stored in anti-static containers at temperatures between 20°C and 40°C and in a humidity-controlled environment of 40-60%rh.

The shelf-life of this product is estimated at 5 years when stored appropriately.

6. MATERIALS AND PROCESSES

6.1 Materials Used

Material	Manufacturer	%TML	%CVC	%WVR	Application	Applicable Products
Araldite 2014 Epoxy	Huntsman	0.97	0.05	0.33	Adhesive fixing	All
1B31 Acrylic	Humiseal	3.89	0.11	0.09	Conformal Coating	All
DC 6-1104	Dow Corning	0.17	0.02	0.06	Adhesive fixing on modifications	All
Stycast 2850	Emerson & Cuming	0.25	0.01	0.05	Adhesive fixing	All
PCB material	FR4	0.62	0	0.1	Note: worst case on NASA out-gassing list	All
Solder Resist	CARAPACE EMP110 or XV501T-4	0.95 or 0.995	0.02 Or 0.001	0.31	-	All
Solder	Sn62 or Sn63 (Tin/Lead)	-	-	-	-	All
Flux	Alpha Rosin Flux, RF800, ROL 0	-	-	-	Low activity flux to avoid corrosion	All
300 Series Stainless Steel	Pemnet	-	-	-	PEMs	01-02453
A4 Stainless Steel (316L)	PTS-UK	-	-	-	M3 Fasteners	01-02453

Table 6-1 Materials List

Part Used	Manufacturer	Contact	Insulator	Type	Use	Required mating Connector	Applicable Products
DF13-6P-1.25DSA(50)	Hirose	Gold Plated	Polyamide	PTH	Programming Header – not for customer use	DF13-6S-1.25C and DF13-2630SCFA(04)	All
DF13-5P-1.25DSA(50)	Hirose	Gold Plated	Polyamide	PTH	Solar Array Connectors	DF13-5S-1.25C and DF13-2630SCFA(04)	All
ESQ-126-39-G-D	Samtec	Gold Plated	Black Glass Filled Polyester	PTH	CubeSat Kit Compatible Headers	ESQ-126 range	All
FTSH-110-01-F-DV	Samtec	Gold Plated Beryllium Copper	Black Liquid Crystal Polymer	SMT	Expansion Header for daughterboard connection	TFM-110-22-L-D-A	25-02451
SFM-110-02-L-D-A	Samtec	Gold Plated Beryllium Copper	Black Liquid Crystal Polymer	SMT	Expansion Header for daughterboard connection	TFM-110-22-L-D-A	25-02452, 01-02453
TFM-110-22-L-D-A	Samtec	Gold Plated Beryllium Copper	Black Liquid Crystal Polymer	SMT	Daughterboard to motherboard connection header	SFM-110-02-L-D-A	01-02453
DF13-8P-1.25DSA(50)	Hirose	Gold Plated	Polyamide	PTH	Solar Array Connectors (Daughter Board)	DF13-8s-1.25C and DF13-2630SCFA(04)	01-02453

Table 6-2 Connector Headers

6.2 Processes and Procedures

All assembly is inspected to ESA Workmanship Standards; ECSS-Q-ST-70-08C and ECSS-Q-ST-70-38C.

7. SYSTEM DESCRIPTION

The Clyde Space EPS products are optimised for Low Earth Orbit (LEO). They are designed for integration with spacecraft that have six body mounted solar panels or fewer (i.e. one on each spacecraft facet), or potentially other configurations involving deployable panels. The EPS can accommodate various solar panel configurations, and has been designed to be versatile; please consult our support team if you have specific requirements for connecting the EPS to your spacecraft.

The Clyde Space EPS connects to the solar panels via a number of independent Battery Charge Regulators (BCRs). These are connected with panels on opposing faces of the satellite connected to the same BCR. Additional BCRs (not applicable to 25-02452) allow deployable panels to be used. In this configuration only one panel per pair can be directly illuminated at any given time, with the second panel providing a limited amount of energy due to albedo illumination. Each of the BCRs has an inbuilt Maximum Power Point Tracker (MPPT). This MPPT will track the dominant panel of the connected pair (the directly illuminated panel).

The output of the BCRs are then connected together and supply charge to the battery, Power Conditioning Modules (PCMs) and Power Distribution Modules (PDMs).

The PCM network has an unregulated Battery Voltage Bus, a regulated 5V supply, a regulated 3.3V supply and a regulated 12V supply, each with a separate Latching Current Limiter (LCL) with automatic retry. In addition to the main buses there are 10 commandable PDMs – 2x12V, 2xBATV, 3x5V and 3x3.3V. The EPS also has multiple inbuilt protection methods to ensure safe operation during the mission and a full range of EPS telemetry via the I²C network. These are discussed in detail in Sections 10 and 11.

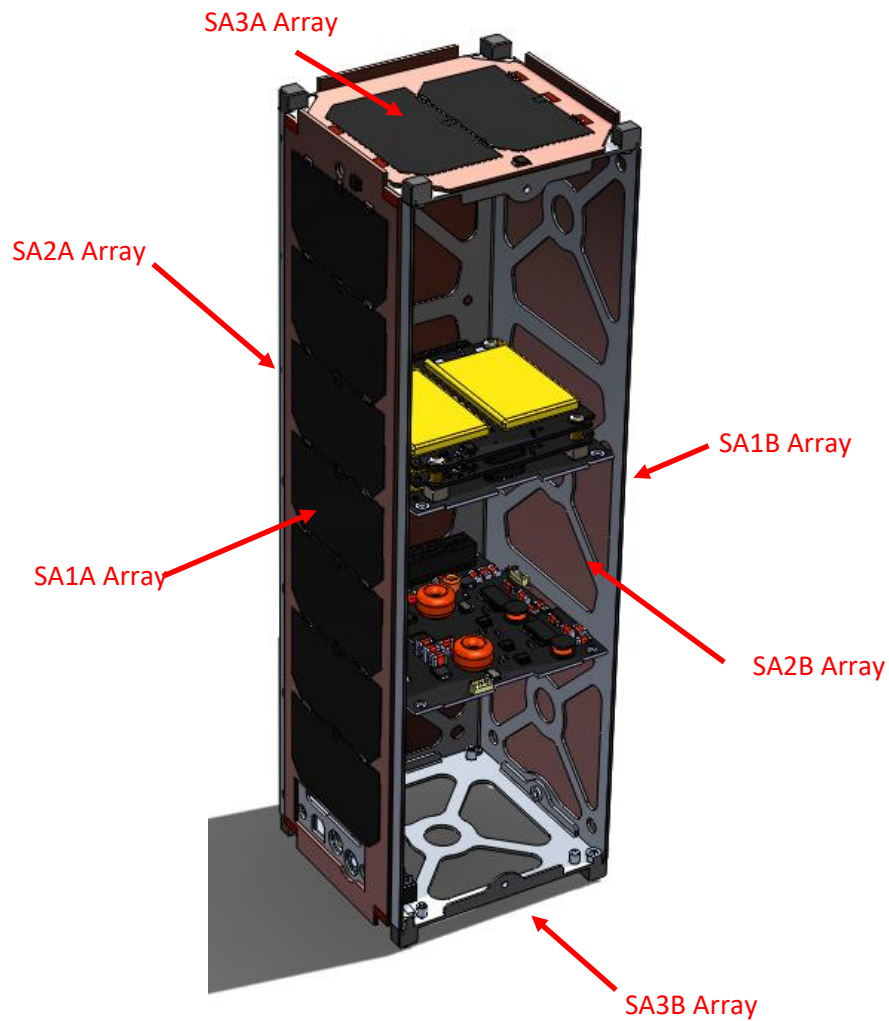


Figure 7-1 Typical Array Configuration for 25-02452 with Example Allocations (for 01-02453, deployed panels can additionally interface to BCRs 4-9)

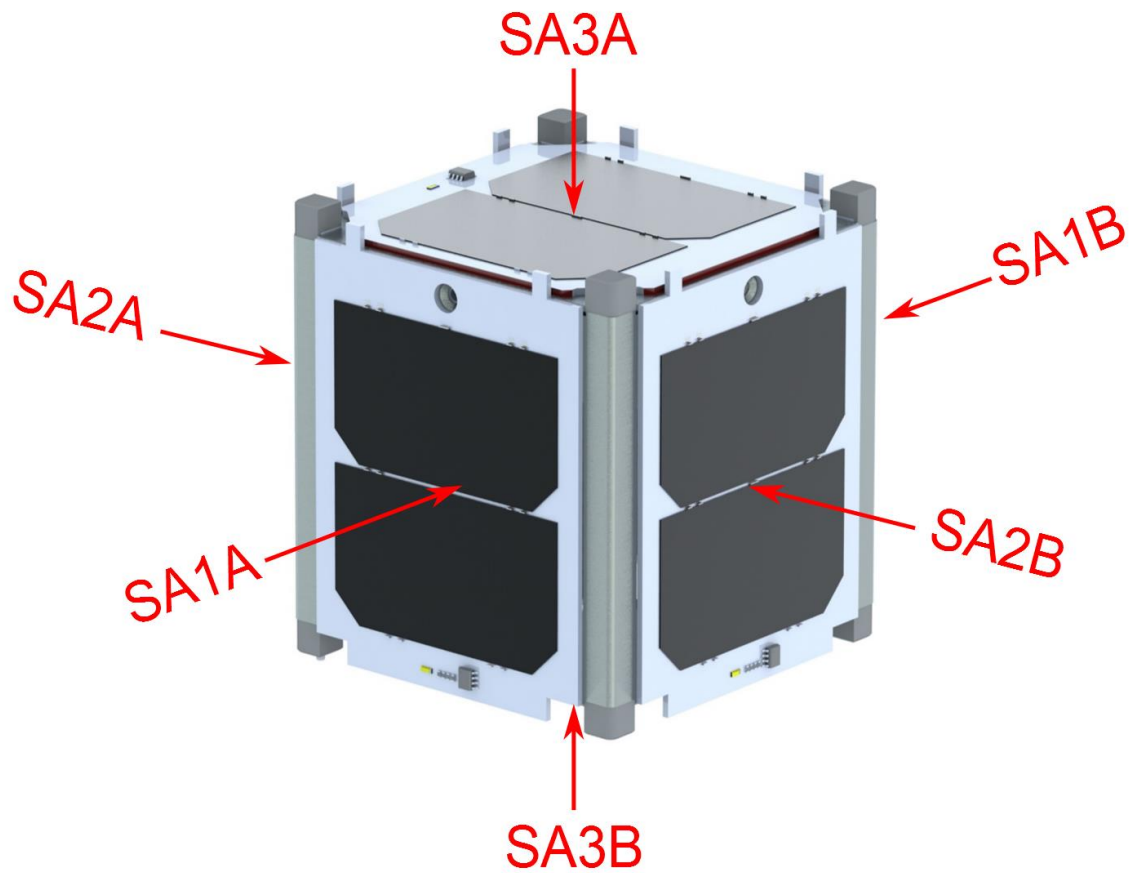


Figure 7-2 Typical Array Configuration for 25-02451 with Example Allocations.
Fourth BCR interface not shown in image.

7.1 System Overview

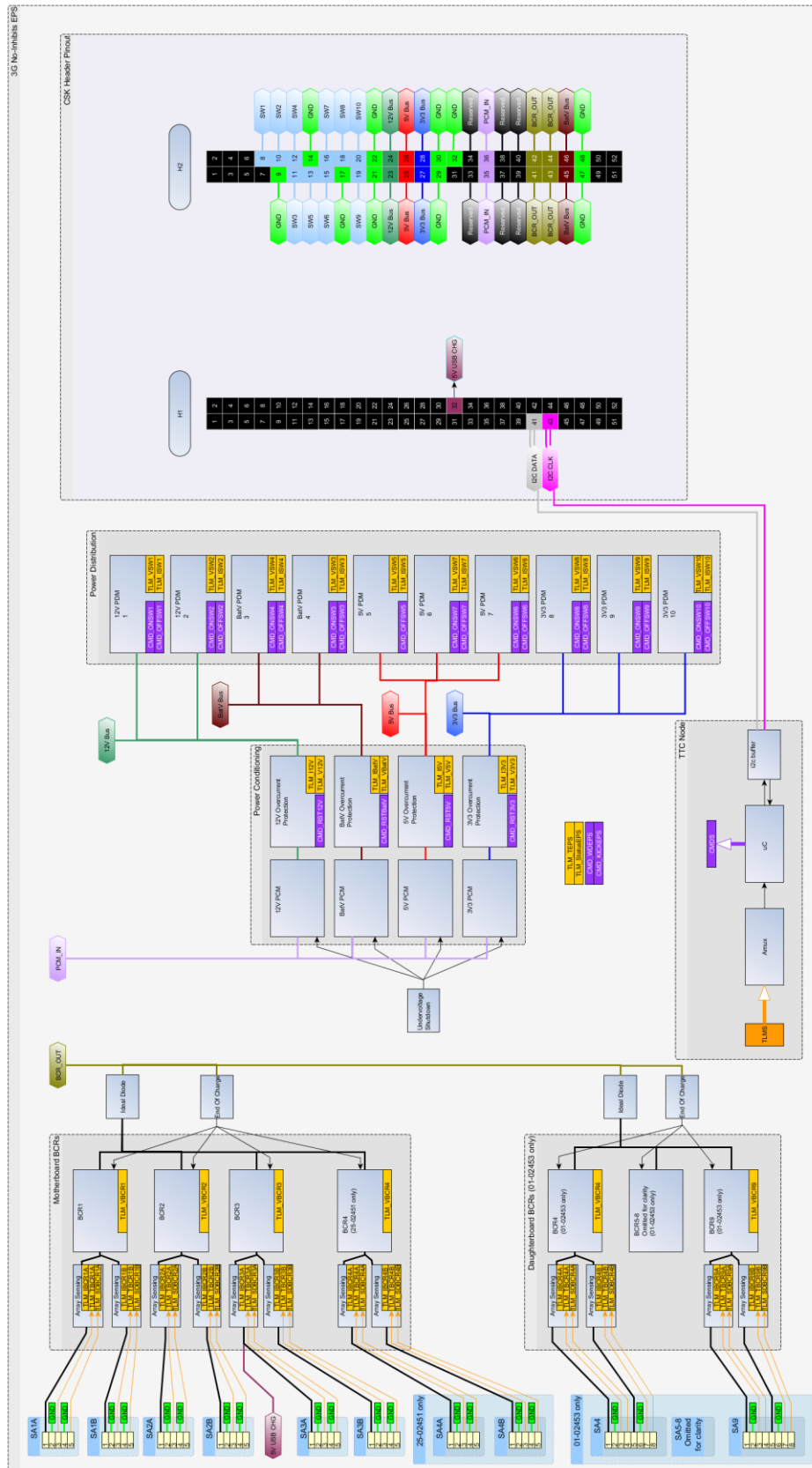


Figure 7-3 Functional Diagram

7.2 Autonomy and Redundancy

All BCR power stages feature full system autonomy, operating solely from the solar array input and not requiring any power from the battery systems. This feature offers graceful degradation of the system as none of the BCRs depend on any other circuitry to operate correctly. Failure of all strings of the battery (any of the Clyde Space battery range) will not damage the BCRs but, due to the MPPT, will result in an intermittent interruption on all power buses (approximately every 2.5 seconds).

The rest of the power system is a robustly designed single string.

7.3 Quiescent Power Consumption

All power system efficiencies detailed (for BCRs and PCMs) take into consideration the associated low level control electronics. As such, these numbers are not included in the quiescent power consumption figures.

The quiescent current draw covers the power required to run the TTC node, PDMs and other monitoring and safety features of the EPS, and values are given in Table 4-1.

7.4 Mass and Mechanical Configuration

The system is contained on a single PC/104 size card, compatible with the Cubesat Kit bus. The 01-02453 FlexU EPS also includes a daughterboard. The 25-02451 1U EPS can be used as the motherboard for a CubeSat power system with an integrated battery. This will be documented in the battery user manual [RD-1]. The 25-02452 and 01-02453 EPS products are expected to be used with a standalone battery.

The masses of the EPS products are specified in Table 7-1 and dimensioned drawings are given in Figure 7-4 through Figure 7-6.

Part number	Min	Typical	Max	Unit
25-02451	84	86	88	g
25-02452	84	86	88	g
01-02453	145	148	150	g

Table 7-1 Mass of EPS Products

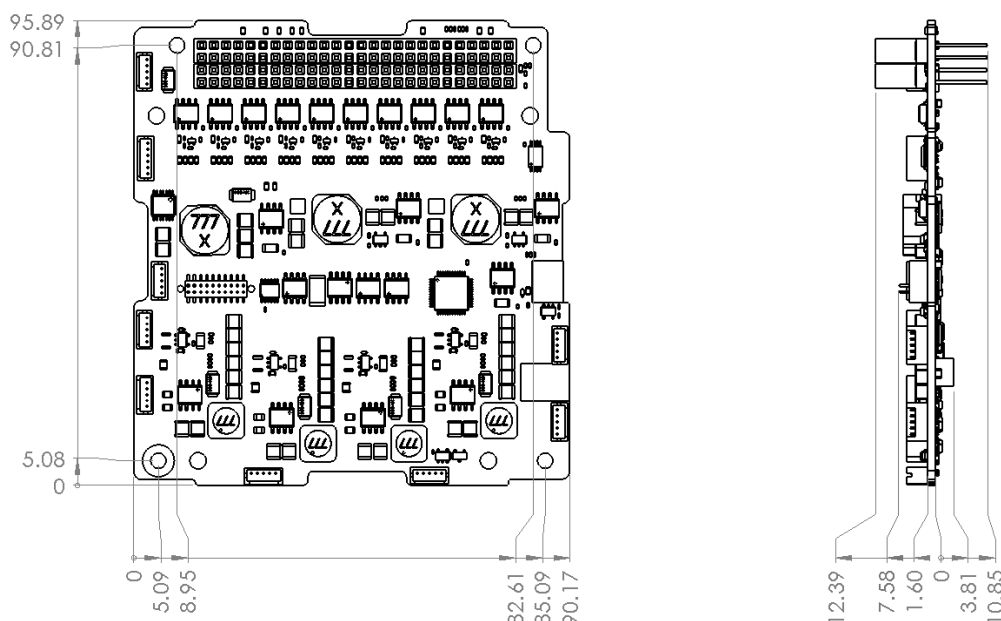


Figure 7-4 Dimensioned Drawing of 25-02451 1UB EPS

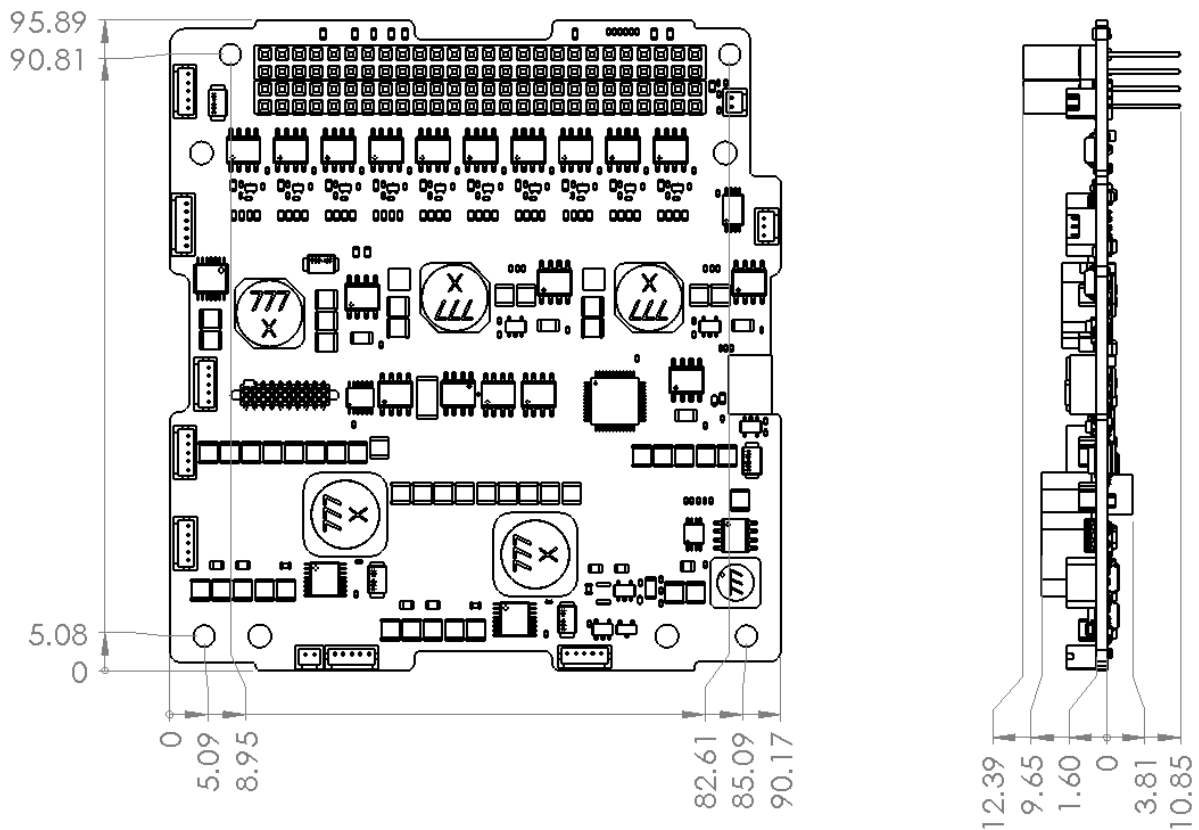


Figure 7-5 Dimensioned Drawing of 25-02452 3UA EPS

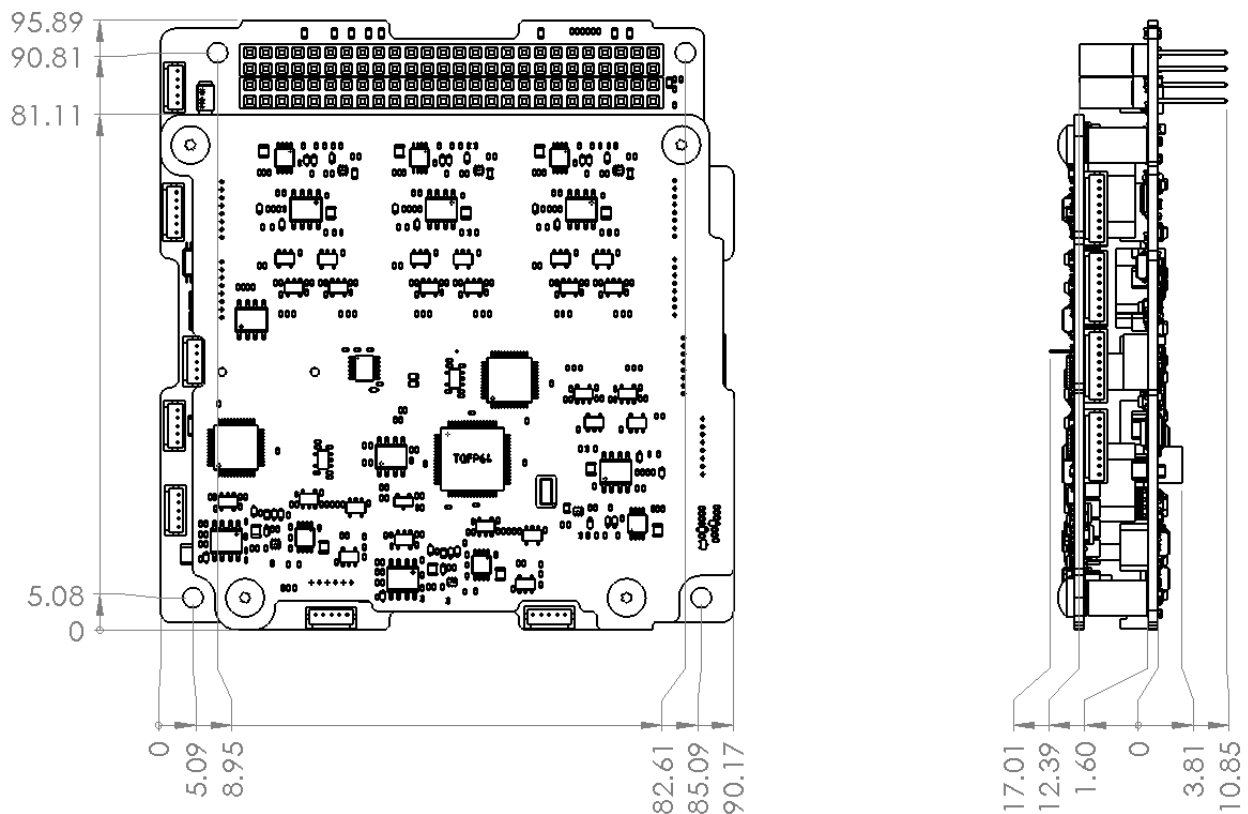


Figure 7-6 Dimensioned Drawing of 01-02453 XUA EPS

8. INTERFACING

The connector interfaces of the EPS are outlined in Figure 8-1, including the solar array inputs, output of the power buses and communication to the I²C node. In the following section, it is assumed that the EPS will be integrated with a Clyde Space Battery.

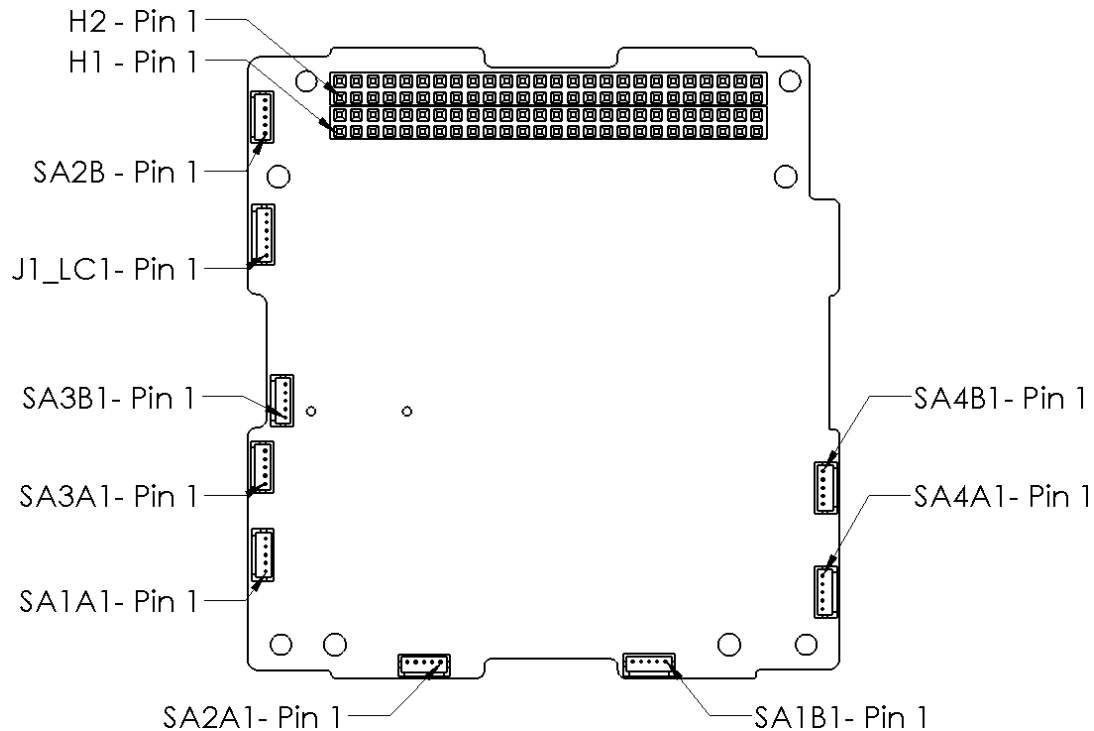


Figure 8-1 Connector Location Diagram for 25-02451 1UB EPS

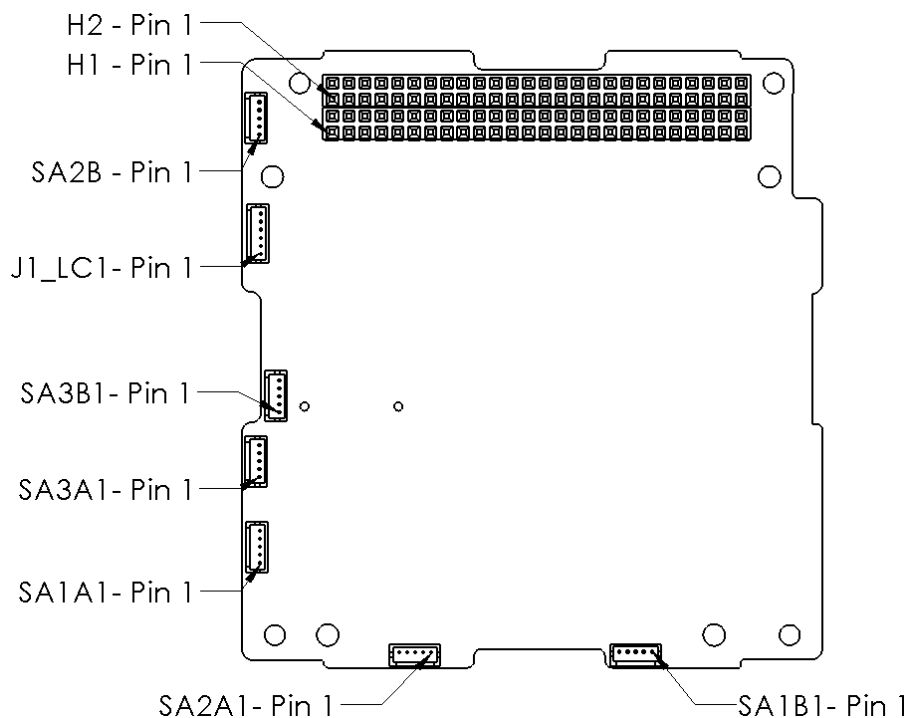
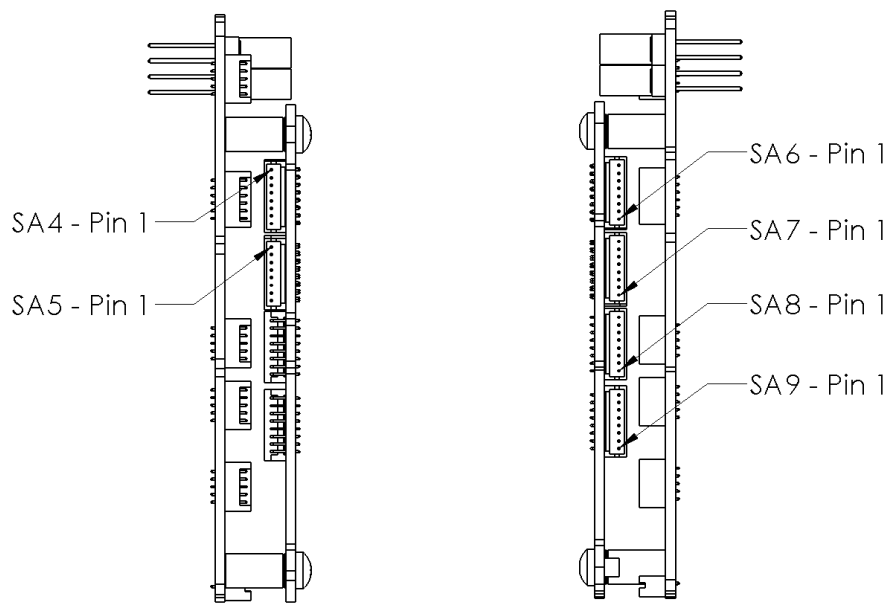


Figure 8-2 Connector Location Diagram for 25-02452 and 01-02453 Motherboard

**Figure 8-3 Additional Daughterboard Connections for 01-02453 XUA EPS**

The connector positions and functions are described in Table 8-1.

Connector	Function	Location	Applicable Products
SA1A	Solar Array connector, BCR1 channel A	Motherboard	All
SA1B	Solar Array connector, BCR1 channel B	Motherboard	All
SA2A	Solar Array connector, BCR2 channel A	Motherboard	All
SA2B	Solar Array connector, BCR2 channel B	Motherboard	All
SA3A	Solar Array connector, BCR3 channel A	Motherboard	All
SA3B	Solar Array connector, BCR3 channel B	Motherboard	All
SA4A	Solar Array connector, BCR4 channel A	Motherboard	25-02451
SA4B	Solar Array connector, BCR4 channel B	Motherboard	25-02451
SA4	Solar Array connector, BCR4 both channels	Daughterboard	01-02453
SA5	Solar Array connector, BCR5 both channels	Daughterboard	01-02453
SA6	Solar Array connector, BCR6 both channels	Daughterboard	01-02453
SA7	Solar Array connector, BCR7 both channels	Daughterboard	01-02453
SA8	Solar Array connector, BCR8 both channels	Daughterboard	01-02453
SA9	Solar Array connector, BCR9 both channels	Daughterboard	01-02453
J1_IC1	Programming header – Clyde Space use only	Motherboard	All
H1	CubeSat Kit bus compatible Header 1	Motherboard	All
H2	CubeSat Kit bus compatible Header 2	Motherboard	All

Table 8-1 Connector functions

8.1 Solar Array Connection

The EPS motherboard has eight (25-02451) or six (25-02452, 01-02453) connectors for the attachment of solar arrays. The EPS daughterboard (01-02453 only) has a further six such connectors. This interface accommodates inputs from the arrays with temperature and sun detector telemetry for each.

HIROSE DP13-5P-1.25DSA(50) connector sockets are used for motherboard solar array inputs. Inputs labelled as A and B are always connected to the BCR in parallel with one another. An example configuration is shown in Figure 8-4. HIROSE DP13-8P-1.25DSA(50) connector sockets are used for daughterboard solar array inputs. An example configuration is shown in Figure 8-5.

All arrays which are connected in parallel should have the same number of cells.

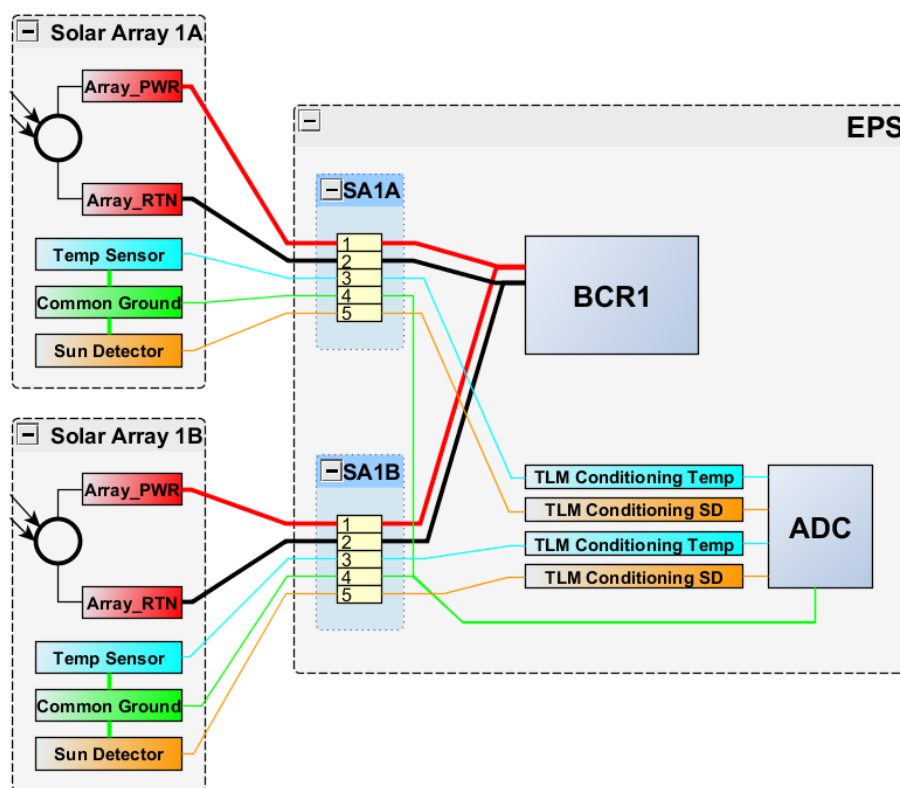


Figure 8-4 Example Solar Array Configuration SA1-3 and (25-02451 only) SA4

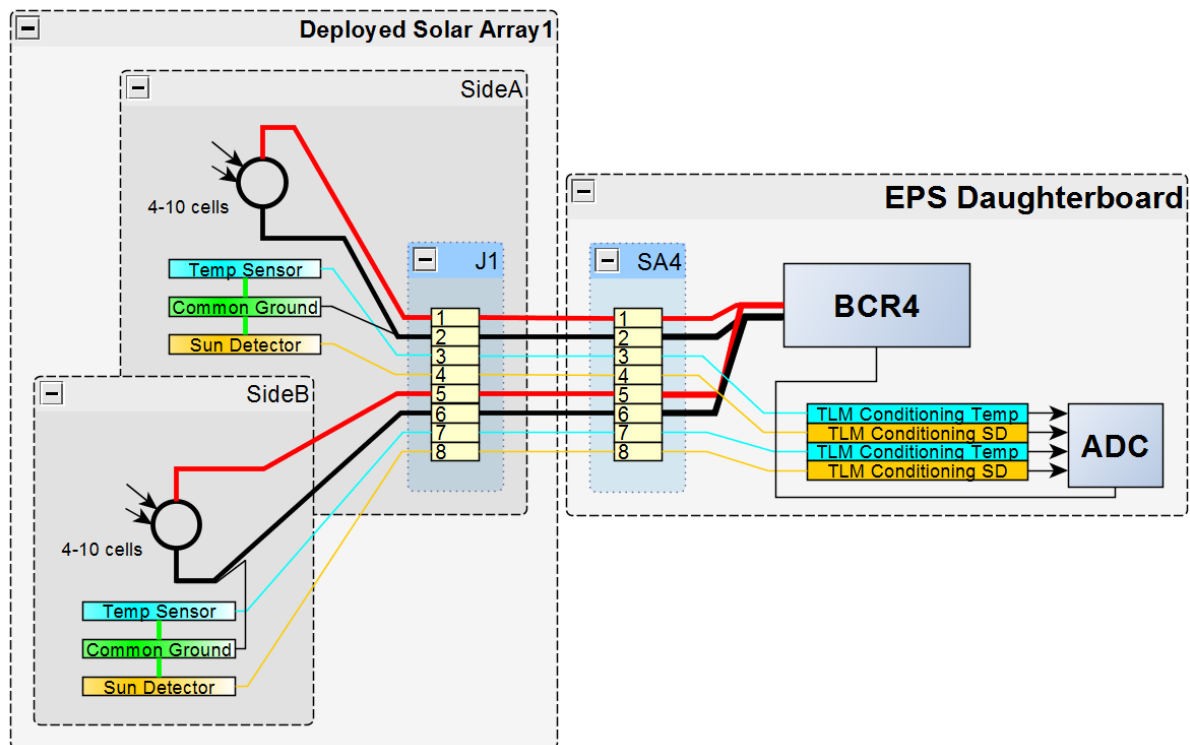


Figure 8-5 Example Solar Array Configuration SA4-9 (01-02453 only)

Pin	Use	Notes
1	Array Power	Connection to positive of solar cell string
2	Array Return	Negative of solar cell string – connected to ground within the EPS
3	Temperature Telemetry	Telemetry
4	Ground Line	Ground connection for Sensors
5	Sun Detector Telemetry	Telemetry

Table 8-2 Pinout for motherboard solar array connectors

Pin	Use	Notes
1	Array A Power	Connection to positive of solar cell string
2	Ground	Negative of solar cell string and ground connection for sensors – connected to ground within the EPS
3	Array A Temperature Telemetry	Telemetry
4	Array A Sun Detector Telemetry	Telemetry
5	Array B Power	Connection to positive of solar cell string
6	Ground	Negative of solar cell string and ground connection for sensors – connected to ground within the EPS
7	Array B Temperature Telemetry	Telemetry
8	Array B Sun Detector Telemetry	Telemetry

Table 8-3 Pinout for daughterboard solar array connectors

8.2 Solar Array Harness

Clyde Space supply harnesses (sold separately) to connect the solar panels to the EPS Motherboard, comprising one Hirose DF13-5S-1.25C connected at the panel and one connector at the other connected at the EPS. Similarly, harnesses to connect the solar panels to the EPS Daughterboard (01-02453 only) comprise one Hirose DF13-8S-1.25C connected at the panel and one connector at the other connected at the EPS.

8.3 Temperature Sensing Interface

A temperature sensor is included on each Clyde Space solar panel and can be connected to the EPS to provide panel temperature telemetry. The output from the sensor is then passed to the telemetry system via on board signal conditioning. The formula for calculating solar array temperature from ADC counts can be found in Section 11.4.

8.4 Sun Detector Interface

A photodiode-based coarse sun detector is included on each Clyde Space solar panel and can be connected to each BCR channel to provide panel illumination telemetry. On-board signal conditioning converts this signal to an ADC count which can be translated into an illumination level using the equations in section 11.4.

8.5 Non-Clyde Space Solar Arrays

When connecting non-Clyde Space solar arrays care must be taken with the polarity. Cells used should be of triple junction type. If other manufacturer's panels are to be interfaced, please contact Clyde Space.

8.6 CubeSat Kit Compatible Headers

Connections from the EPS to the bus of the satellite are made via the CubeSat Kit compatible headers H1 and H2, as shown in Figure 8-6.

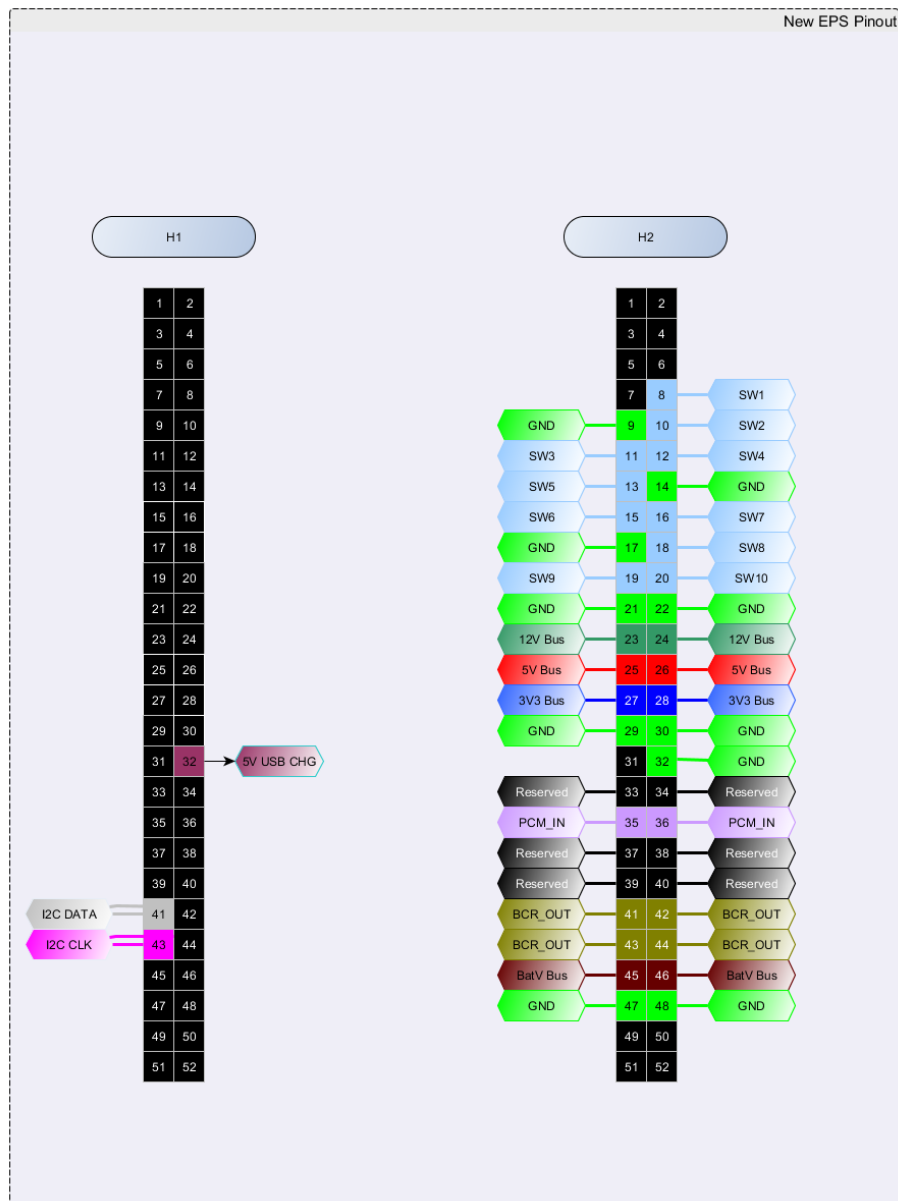


Figure 8-6 CubeSat Kit Header Schematic

8.7 Cubesat Kit Header Pin Out

HEADER 1				HEADER 2			
Pin	Name	Use	Notes	Pin	Name	Use	Notes
1	NC	-	-	1	NC	-	-
2	NC	-	-	2	NC	-	-
3	NC	-	-	3	NC	-	-
4	NC	-	-	4	NC	-	-
5	NC	-	-	5	NC	-	-
6	NC	-	-	6	NC	-	-
7	NC	-	-	7	NC	-	-
8	NC	-	-	8	SW1	PDM 1 Output	12V PDM
9	NC	-	-	9	GND	Ground	System Ground
10	NC	-	-	10	SW2	PDM 2 Output	12V PDM
11	NC	-	-	11	SW3	PDM 3 Output	BAT PDM
12	NC	-	-	12	SW4	PDM 4 Output	BAT PDM
13	NC	-	-	13	SW5	PDM 5 Output	5V PDM
14	NC	-	-	14	GND	Ground	System Ground
15	NC	-	-	15	SW6	PDM 6 Output	5V PDM
16	NC	-	-	16	SW7	PDM 7 Output	5V PDM
17	NC	-	-	17	GND	Ground	System Ground
18	NC	-	-	18	SW8	PDM 8 Output	3V3 PDM
19	NC	-	-	19	SW9	PDM 9 Output	3V3 PDM
20	NC	-	-	20	SW10	PDM 10 Output	3V3 PDM
21	NC	-	-	21	GND	Ground	System Ground
22	NC	-	-	22	GND	Ground	System Ground
23	NC	-	-	23	12VBUS	12V Bus	Power Bus
24	NC	-	-	24	12VBUS	12V Bus	Power Bus
25	NC	-	-	25	5VBUS	5V Bus	Power Bus
26	NC	-	-	26	5VBUS	5V Bus	Power Bus
27	NC	-	-	27	3V3BUS	3.3V Bus	Power Bus
28	NC	-	-	28	3V3BUS	3.3V Bus	Power Bus
29	NC	-	-	29	GND	Ground	System Ground
30	NC	-	-	30	GND	Ground	System Ground
31	NC	-	-	31	NC	-	-
32	5VUSB_CHG	5V USB Charge	Battery Top up Charge	32	GND	Ground	System Ground
33	NC	-	-	33	Reserved	Do not use	Reserved
34	NC	-	-	34	Reserved	Do not use	Reserved
35	NC	-	-	35	PCM_IN	PCM Input	Reserved
36	NC	-	-	36	PCM_IN	PCM Input	Reserved
37	NC	-	-	37	Reserved	Do not use	Reserved
38	NC	-	-	38	Reserved	Do not use	Reserved
39	NC	-	-	39	Reserved	Do not use	Reserved
40	NC	-	-	40	Reserved	Do not use	Reserved
41	I2C_DATA	I2C Data	-	41	BCR_OUT	BCR Output	Reserved
42	NC	-	-	42	BCR_OUT	BCR Output	Reserved
43	I2C_CLK	I2C Clock	-	43	BCR_OUT	BCR Output	Reserved
44	NC	-	-	44	BCR_OUT	BCR Output	Reserved
45	NC	-	-	45	BatVBUS	Unregulated Battery Bus	Power Bus
46	NC	-	-	46	BatVBUS	Unregulated Battery Bus	Power Bus
47	NC	-	-	47	GND	Ground	System Ground
48	NC	-	-	48	GND	Ground	System Ground
49	NC	-	-	49	NC	-	-
50	NC	-	-	50	NC	-	-
51	NC	-	-	51	NC	-	-
52	NC	-	-	52	NC	-	-

Table 8-4 Pin Descriptions for Header H1 and H2

8.8 Flight Switches

The Flight Switches provide a method of isolating the BCRs and battery from the satellite power buses during storage, transportation and launch.

This EPS does not have flight switches populated as it is designed to be used with a compatible battery which contains the inhibits. Refer to **User manual: 3rd Generation CubeSat Battery Family [RD-1]** for information on flight switches, Sections *10.1 Protection Overview* and *10.7 Inhibit Operation*.

8.9 Battery connection

Connection of the battery systems on the EPS is via the CubeSat kit bus or via an integrated battery (25-02451 only). Ensure that the pins are aligned, and located in the correct position, as any offset can cause the battery to be shorted to ground, leading to catastrophic failure of the battery and damage to the EPS. It is also important that the EPS is only used with compatible battery products; see Section 13.1 for information.

Failure to observe these precautions will result in the voiding of any warranty.

When a battery board is connected to the CubeSat Kit header and the battery inhibits are not activated, there are live unprotected battery pins accessible (H2.35-36 and H2.41-44). These pins should not be routed to any connections other than the Clyde Space EPS, otherwise all EPS-based protections will be bypassed and significant battery damage can be sustained.

9. TECHNICAL DESCRIPTION

This section gives a complete overview of the operational modes of the EPS. It is assumed that a complete Clyde Space power system (EPS, Batteries and Solar panels) is in operation for the following sections.

9.1 Charge Method

The BCR charging system has two modes of operation: Maximum Power Point Tracking (MPPT) mode and End of Charge (EoC) mode. These modes are governed by the state of charge of the battery.

MPPT Mode

If the battery voltage is below the EoC voltage the system is in MPPT mode. This is based on constant current charge method, operating at the maximum power point of the solar panel for maximum power transfer.

EoC Mode

Once the EoC voltage has been reached, the BCR changes to EoC mode, which is a constant voltage charging regime. The EoC voltage is held constant and a tapering current from the panels is supplied to top up the battery until at full capacity. In EoC mode the MPPT circuitry moves the solar array operation point away from the maximum power point of the array, drawing only the required power from the panels. The excess power is left on the arrays as heat, which is transferred to the structure via the array's thermal dissipation methods incorporated in Clyde Space panels.

The operation of these two modes can be seen in Figure 9-1.

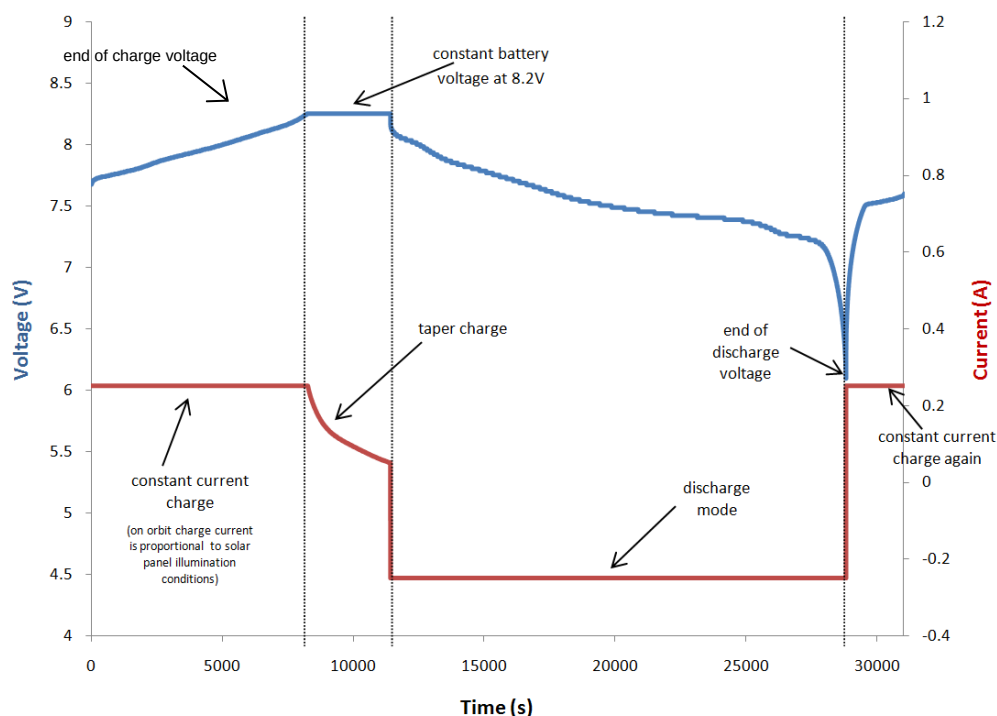


Figure 9-1 Tapered charging method

The application of constant current/constant voltage charge method on a spacecraft is described in more detail in RD-6. In this document, there is on-orbit data showing the operation and how the current fluctuates with changing illumination conditions and orientation of the spacecraft with respect to the Sun.

9.2 BCR Power Stage Overview

The EPS has several separate, independent BCRs, each designed to interface to two parallel solar arrays on opposing faces of the satellite.

Each design offers a highly reliable system that can deliver 90% (Buck BCRs) or 80% (SEPIC BCRs) of the power delivered from the solar array network at full load.

BCR Allocation

The EPS has Buck and SEPIC BCRs as listed below.

BCR Type	25-02451	25-02452	01-02453
Buck BCR	N/A	1, 2	1, 2, 4, 5, 6, 7, 8, 9
SEPIC BCR	1,2,3,4	3	3

Buck BCR Power Stage

The Buck BCRs allow the EPS to interface to strings of four to eight cells in series. This will deliver up to 90% output at full load. The design will operate with input voltages between 7.4V and 30V. If the maximum power point is below 9.4V, the MPPT will drift away from the maximum power point of the array at end of charge, sacrificing power system efficiency.

SEPIC BCR Power Stage

The SEPIC BCRs allow the EPS to interface to solar arrays of two triple junction cells in series. This will deliver up to 80% output at full load. The BCR will operate with an input of between 3.0V and 9.5V.

9.3 MPPT

Each of the BCRs can have two solar arrays connected at any given time; only one array can be illuminated by sunlight, although the other may receive illumination by albedo reflection from earth. The dominant array is in sunlight and this will operate the MPPT for that BCR string. The MPPT monitors the power supplied from the solar array. This measurement is used to calculate the maximum power point of the array. The system tracks this point by periodically adjusting the BCRs to maintain the maximum power derived from the arrays. This technique ensures that the solar arrays can deliver much greater usable power, increasing the overall system performance.

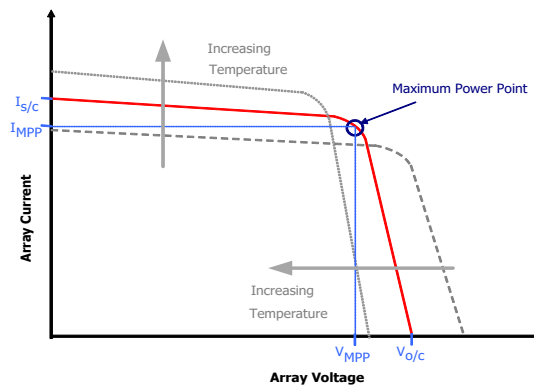


Figure 9-2 Solar Array Maximum Power Point

The monitoring of the MPP is done approximately every 2.5 seconds. During this tracking, the input of the array will step to o/c voltage, as shown in Figure 9-3.

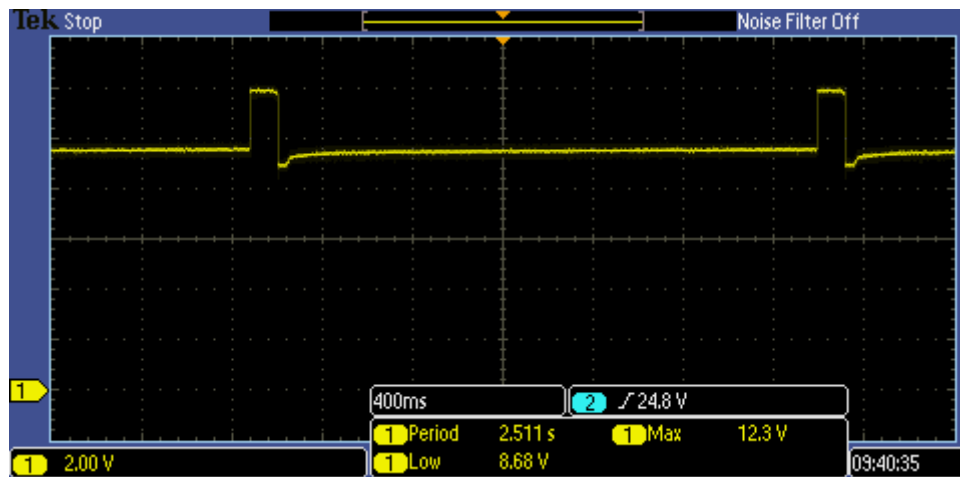


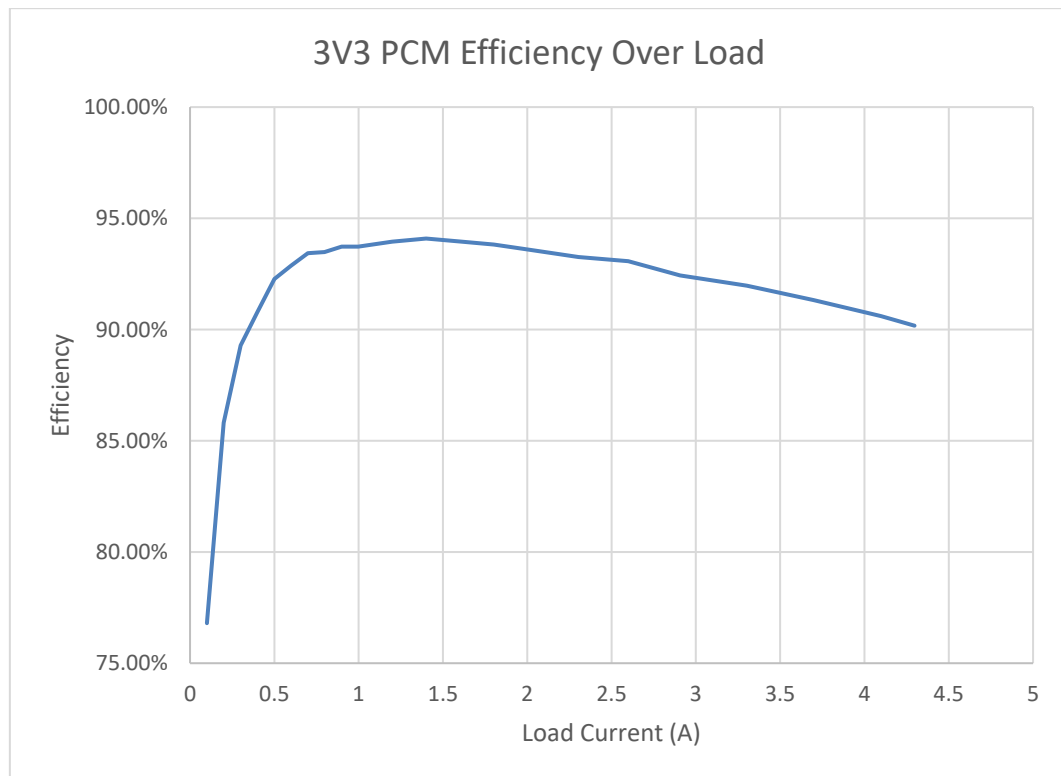
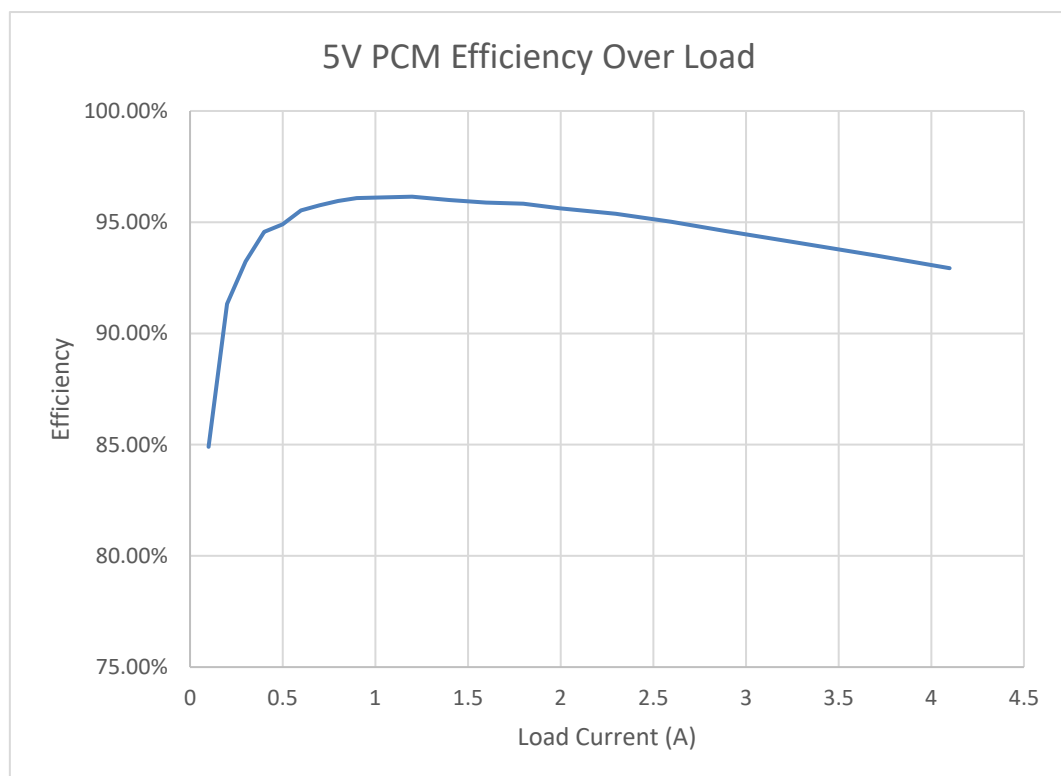
Figure 9-3 Input waveform with Maximum Power Point Tracking

9.4 5V USB Charge

The EPS offers a method of trickle charging the battery by connecting a power supply directly to the 5V USB Charge pin on the header. The 5V USB Charge will operate with an input of between 3.0V and 6V. This charge connection utilises BCR3 to charge the battery and provides a parallel input to the 3W BCR. It should be noted that the 5V USB is designed for trickle charging the battery and as such will provide a maximum of 2.1W when operated at 6V input.

9.5 5V and 3.3V PCMs with Latching Current Limiter

The 5V and 3.3V regulators both use buck switching topology regulators as their main converter stage. The regulator maintains the output voltage within +/- 1% of nominal. Efficiency curves are given in Figure 9-4 and Figure 9-5. The current limit of each regulator is nominally 4.5A. Each regulator operates at a frequency of 480 kHz. The Latching Current Limiter is described in Section 9.8. If an over-current event triggers the Latching Current Limiter a retry circuit will attempt to re-enable the bus as described in Section 10.1.

**Figure 9-4 Efficiency Curve for 3V3 PCM, Vbat=7.6v, Tbrd=23°C****Figure 9-5 Efficiency Curve for 5V PCM, Vbat=7.6v, Tbrd=23°C**

9.6 12V PCM with Latching Current Limiter

The 12V regulator uses a boost switching topology regulator as the main converter stage. The regulator maintains the output voltage within $\pm 1\%$ of nominal. Efficiency is plotted in Figure 9-6. The current limit on the regulator is nominally 1.5A. The regulator operates at a frequency of 800 kHz. The Latching Current Limiter is described in Section 9.8. If an over-current event triggers the Latching Current Limiter, a retry circuit will attempt to re-enable the bus as described in Section 10.1.

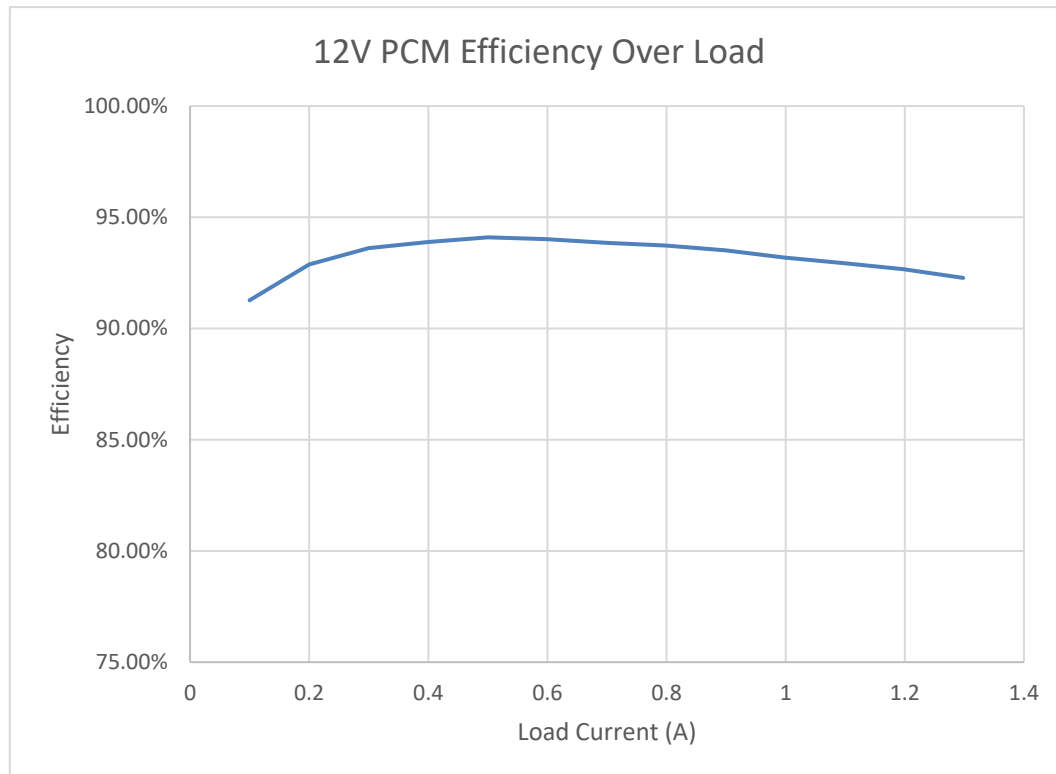


Figure 9-6 Efficiency Curve for 12V PCM, $V_{bat}=7.6V$, $T_{brd}=23^{\circ}C$

9.7 BatV PCM with Latching Current Limiter

The unregulated battery voltage regulator provides safe access to the battery bus of the satellite. The voltage supplied will vary directly with the battery voltage (between 6.144V and 8.26V). The current limit is nominally 4.7A. The Latching Current Limiter is described in Section 9.8. If an over-current event triggers the Latching Current Limiter, a retry circuit will attempt to re-enable the bus as described in Section 10.1.

9.8 PDMs with Latching Current Limiter

Ten independently commandable power distribution modules (PDM) are included on the EPS. Each PDM has inbuilt overcurrent protection in the form of a latching current limiter (LCL). By utilising an LCL each PDM is capable of driving loads with large inrush currents without compromising safety throughout the duration of the mission (this is of particular interest for applications such as transceivers). Once the LCL has activated, turning off the supply of power, the PDM will remain off until commanded to switch on again. The PDMs cover the range of regulated and unregulated voltages provided by the EPS.

LCL Operation Description

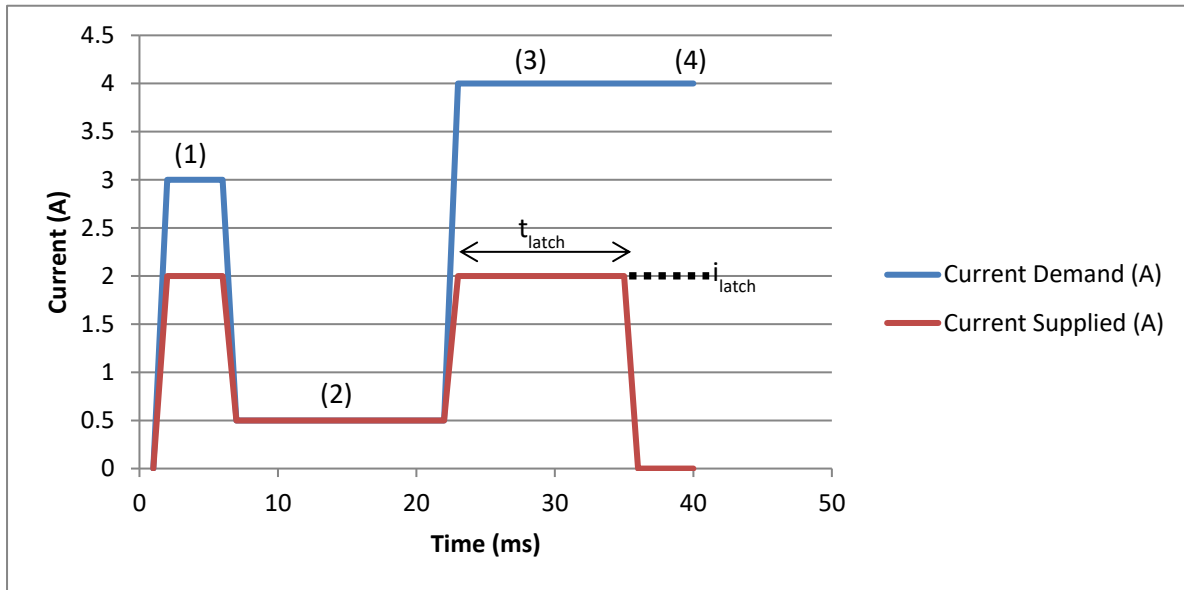


Figure 9-7 Latching Current Limiter Example Operation

In the example system shown above the events are as follows:

1. The payload demands a 3A initial current, however the PDM limits the current to 2A. The time this demand is present is less than the latch time of the PDM (t_{latch}), so the PDM does not switch off.
2. The payload demand drops to 0.5A. This is below the current limit of the PDM (i_{latch}).
3. A fault condition occurs resulting in a demand of 4A. The PDM only allows 2A to pass, preventing high current damage to the PDM or the payload.
4. The fault remains for longer than t_{latch} so the PDM turns off preventing any current flow.

LCL Characteristics

The following characteristics are specified at 25°C:

- i_{latch} : The latching current limit is set to allow the maximum safe current the EPS can deliver. This value has been selected based on the fact that, if the current limit is set high to allow a high inrush it will result in a high current limit during normal operation too.
- t_{latch} : The latching has been set to allow for the maximum safe length of time before shutting down the bus, allowing capacitive loads to be charged safely.
- C_{latch} : This is the maximum capacitance that can be charged via the LCL before the PDM automatically disables.
- t_{on} : Time delay from PDM being commanded to turn on via I2C node to actual PDM turn on.

PDM#	Pin	Voltage (V)	i_{latch} (A)	t_{latch} (ms)	C_{latch} (μF)	t_{on} (ms)
1	H2.08	12	1 - 1.1	2-3	240	0.140
2	H2.10	12	1 - 1.1	2-3	240	0.140
3	H2.11	BAT	4.2 - 4.3	8-9	800	0.140
4	H2.13	BAT	4.2 - 4.3	8-9	800	0.140
5	H2.13	5	1 - 1.1	8-9	1600	0.140
6	H2.15	5	1 - 1.1	8-9	1600	0.140
7	H2.16	5	1 - 1.1	8-9	1600	0.140
8	H2.18	3.3	1 - 1.1	13-14	4000	0.140
9	H2.19	3.3	1 - 1.1	13-14	4000	0.140
10	H2.20	3.3	1 - 1.1	13-14	4000	0.140

Table 9-1 PDM Switch Configuration

10. GENERAL PROTECTION

The EPS (and wider power system) has a number of inbuilt protections and safety features designed to maintain safe operation of the EPS, battery and all subsystems supplied by the EPS buses.

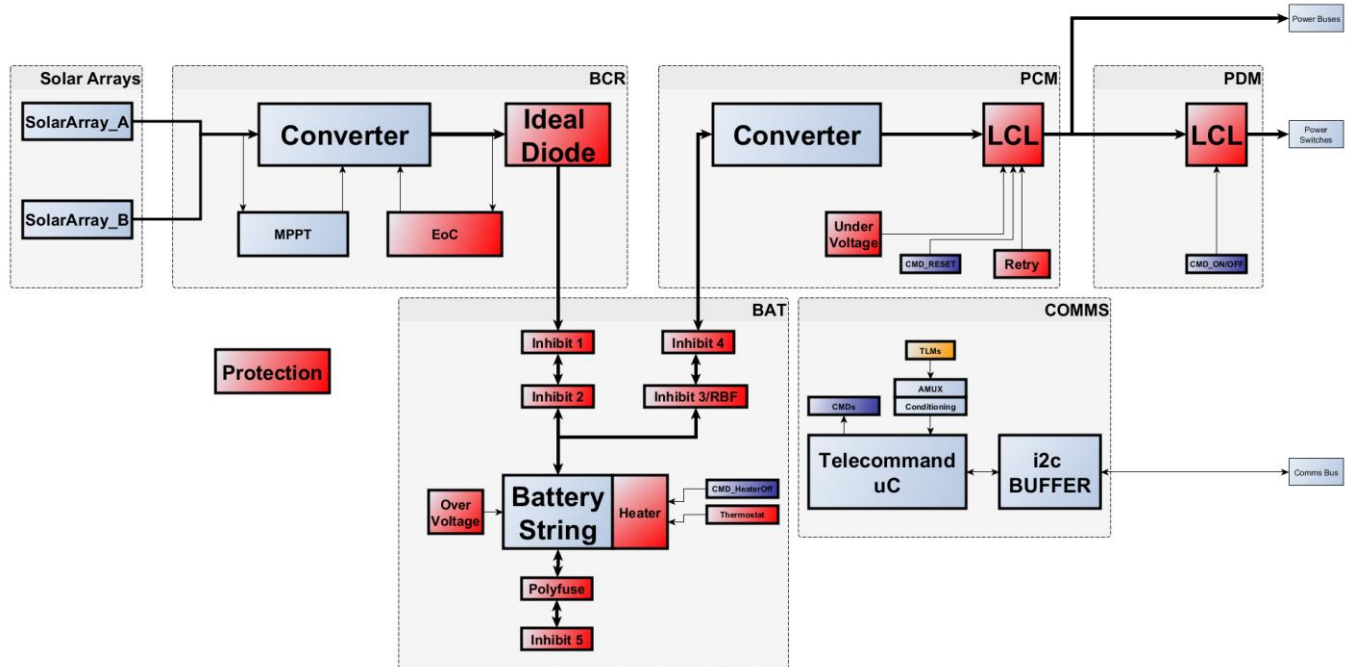


Figure 10-1 Protection Systems (When used with Manned Flight battery)

10.1 Over-Current Bus Protection (LCL)

The EPS features bus protection systems to safeguard the battery, EPS and attached satellite sub-systems. This is achieved using current monitors and a shutdown network within the PCMs.

Over-current shutdowns are present on all buses for sub system protection. These are solid state switches that monitor the current and shut down at predetermined load levels. The bus protection will then monitor the fault periodically and reset when the fault clears. The fault detection and clear is illustrated in the waveform in Figure 10-2.

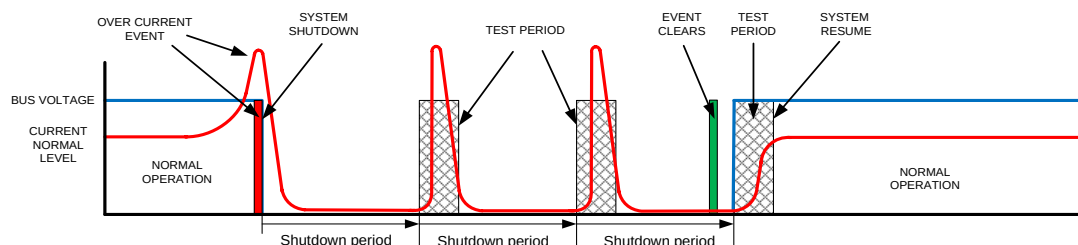


Figure 10-2 Current protection system diagram

The length of time of the test period will depend on the demand caused by the fault condition. Higher current demand results in a shorter test period. All PDMs and buses are protected against a short circuit fault.

10.2 Battery Under-Voltage Protection

In order to prevent the over-discharge of the battery, the EPS has in-built under-voltage shutdown. This is controlled by a comparator circuit with hysteresis. In the event of the battery discharging to ~6.144V (slightly above the level that results in significant battery degradation) the EPS will shut down the supply buses. This will also result in the I²C node shutting down. When a power source is applied to the EPS (e.g. an illuminated solar panel) the battery will begin charging immediately. The buses, however, will not reactivate until the battery voltage has risen to ~7V. This allows the battery to charge to a level capable of sustaining the power lines once a load is applied.

It is recommended that the battery state of charge is monitored and loading adjusted appropriately (turning off of non-critical systems) when the battery capacity is approaching the lower limit. This will prevent the hard shutdown provided by the EPS.

Once the under-voltage protection is activated there is a monitoring circuit used to monitor the voltage of the battery. This will draw approximately 2mA for the duration of shutdown. As the EPS is designed for low earth orbit, the maximum expected period in under-voltage is estimated to be approximately 40mins – after this time, the illuminated panels should bring the battery back above the 7V switch-on voltage. When ground testing this should be taken into consideration, and the battery should be recharged as soon as possible after reaching under-voltage, otherwise permanent damage may be sustained.

11. TELEMETRY AND TELECOMMAND

The telemetry node allows the satellite on board computer (OBC) to monitor the operation of the EPS, control switchable buses and reset the power supplies if this is required for payload or platform recovery operations.

The telemetry node consists of a microcontroller which interfaces to the various telemetry sensing circuits on the EPS through an analogue multiplexer and ADC. The microcontroller is configured to connect through a buffer circuit to the I²C bus of the satellite as a slave node. In response to I²C telemetry requests the microcontroller will sample the desired channel and allowing it to be read back over the I²C bus. In response to a telecommand, the telemetry node will decode the incoming message and reset the desired power bus.

11.1 Communications

All communications to the Telemetry and Telecommand (TTC) node are made using an I²C interface which is configured as a slave and only responds to direct commands from a master I²C node - no unsolicited telemetry is transmitted. The 7-bit I²C address of the TTC node is factory set at 0x2B and the I²C node will operate at a 100kHz bus clock.

Command Protocol

Two message structures are available to the master; a write command and a read command. The write command is used to initiate an event and the read command returns the result. All commands start with the 7-bit slave address and are followed by the data bytes. When reading responses, all data bytes should be read out together. Each command has a delay associated with it, this is required to allow the microcontroller time to process each request. During this delay, the correct response may not be returned, and commands sent during the period may be ignored.

For a write command the first data byte will determine the command to be initiated. The second byte contains the parameters associated with that command. For commands which have no specific requirement for a parameter the second data byte should be set to 0x00.

For a read command, the first data byte represents the most significant byte of the result and the second data byte represents the least significant byte.

Before sending a command, the master is required to set a start condition on the I²C bus. Between each byte the receiving device is required to acknowledge receipt of the previous byte in accordance with the I²C protocol. This will often be accommodated within the driver hardware or software of the I²C master however the user should ensure that this is the case.

The read and write command definitions are illustrated in Figure 11-1.

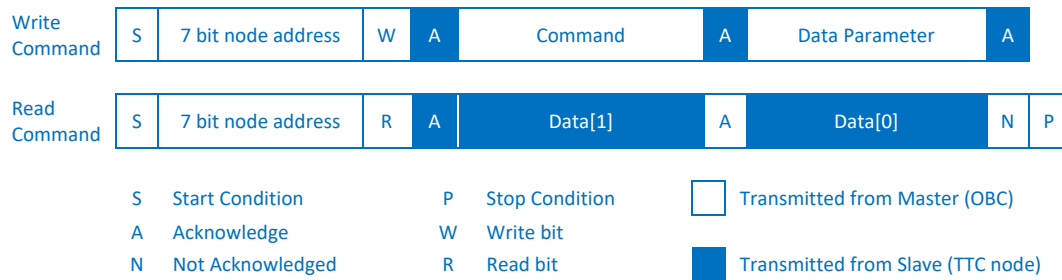


Figure 11-1 I2C Write and Read of 2 byte command packet

If an error has been generated from a command, then the return value will be 0xFFFF. If this value is returned, it is recommended to either inspect the status bytes or to request the code representing the last error generated on the board as described in Section 11.3.

11.2 List of Available Commands

Name	Command	Data[1] ²	Data[0]	25-02451, 25-02452		01-02453	
				Bytes Returned	W/R Delay	Bytes Returned	W/R Delay
Board Status	0x01	NA	0x00	2	1	4	2
Get Last Error	0x03	NA	0x00	2	1	4	2
Get Version	0x04	NA	0x00	2	1	4	2
Get Checksum	0x05	NA	0x00	2	35	4	70
Get Revision	0x06	NA	0x00	2	1	4	2
Get Telemetry	0x10	Table 11-8		2	5	2	15
Get Communications Watchdog Period	0x20	NA	0x00	2	1	2	1
Set Communications Watchdog Period	0x21	NA	Period	0	-	0	-
Reset Communications Watchdog	0x22	NA	0x00	0	-	0	-
Get Number of Brown-out Resets	0x31	NA	0x00	2	1	4	2
Get Number of Auto Software Resets	0x32	NA	0x00	2	1	4	2
Get Number of Manual Resets	0x33	NA	0x00	2	1	4	2
Get Number of Comms Watchdog Resets	0x34	NA	0x00	2	1	2	1
Switch On All PDMs	0x40	NA	0x00	0	-	0	-
Switch Off All PDMs	0x41	NA	0x00	0	-	0	-
Get Actual State of All PDMs	0x42	NA	0x00	4	20	4	20
Get Expected State of All PDMs	0x43	NA	0x00	4	1	4	1
Get Initial State of All PDMs	0x44	NA	0x00	4	20	4	20
Set All PDMs to Initial State	0x45	NA	0x00	4	20	4	20
Switch PDM-N "On"	0x50	NA	N	0	-	0	-
Switch PDM-N "Off"	0x51	NA	N	0	-	0	-
Set PDM-N's Initial State to "On"	0x52	NA	N	0	200	0	200
Set PDM-N's Initial State to "Off"	0x53	NA	N	0	200	0	200
Get PDM-N's Actual Status	0x54	NA	N	2	2	2	2
Set PDM-N's Timer Limit	0x60	N	Limit	0	200	0	150
Get PDM-N's Timer Limit	0x61	NA	N	0	5	0	5
Get PDM-N's Current Timer Value	0x62	NA	N	0	1	0	1
PCM Reset	0x70	NA	Table 11-14	0	1	0	1
Manual Reset	0x80	NA	0x00	0	-	0	-

² Where a command has Data[1] listed as NA, the command only requires a single data byte to be transmitted. This will be given by Data[0].

11.3 Housekeeping and Status Commands

Board Status (0x01)

	Command	Data[0]	Bytes Returned	Delay, ms
25-02451, 25-02452	0x01	0x00	2	1
01-02453	0x01	0x00	4	2

The status bytes are designed to supply operational data about the I²C Node. To retrieve the data that represent the status, the command **0x01** should be sent followed by **0x00**. The meaning of each bit of the returned status bytes is shown below. Please note that Data[1] is the first byte returned from the EPS and Data[0] is the last, this is shown in detail by Figure 11-1. The first two bytes returned represent the status of the motherboard and, in the case of 01-02453, a further two bytes are returned to reflect the status for the daughterboard.

Data[n]	Bit	Description
0	0	Set HIGH if last command not recognised
	1	Set HIGH if a watchdog error occurred, resetting the device
	2	Set HIGH if the data sent along with the last command was incorrect
	3	Set HIGH if the channel passed with the last command was incorrect
	4	Set HIGH if there has been an error reading the EEPROM
	5	Set HIGH if a Power On Reset error occurred
	6	Set HIGH if a Brown Out Reset occurred
	7	Unused
1	0	Unused
	...	
	7	

Table 11-2 Status bits for EPS

Get Last Error (0x03)

	Command	Data[0]	Bytes Returned	Delay, ms
25-02451, 25-02452	0x03	0x00	2	1
01-02453	0x03	0x00	4	2

If an error has been generated after attempting to execute a user's command the value 0xFFFF is returned. To retrieve the details of the last error, send the command **0x03** followed by the data byte **0x00**. This will return the code of the last error generated. Details of each error code are given by Table 11-3. The first two bytes returned represent the Motherboard's error code and, in the case of 01-02453, the second two bytes represent the Daughterboard's.

Code	Description
0x10	CRC code does not match data
0x01	Unknown command received
0x02	Supplied data incorrect when processing command
0x03	Selected channel does not exist
0x04	Selected channel is currently inactive
0x13	A reset had to occur
0x14	There was an error with the ADC acquisition
0x20	Reading from EEPROM generated an error
0x30	Generic warning about an error on the internal SPI bus (only if daughterboard is connected)

Table 11-3 List of Clyde Space Error Codes

Get Version (0x04)

	Command	Data[0]	Bytes Returned	Delay, ms
25-02451, 25-02452	0x04	0x00	2	1
01-02453	0x04	0x00	4	2

The version number of the firmware will be returned on this command.

	Data[1]										Data[0]					
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Value	Board Firmware Number															

Table 11-4 Returned Firmware Version Number

Get Checksum (0x05)

	Command	Data[0]	Bytes Returned	Delay, ms
25-02451, 25-02452	0x05	0x00	2	35
01-02453	0x05	0x00	4	70

This command instructs the node to self-inspect its ROM contents in order to generate a checksum. The value retrieved can be used to determine whether the contents of the ROM have changed during the operation of the device. The first two bytes returned represent the motherboard checksum and, in the case of 01-02453, a further two bytes are returned to for the daughterboard checksum.

Bit	Data[1]								Data[0]							
	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
Value	Board Checksum															

Get Firmware Revision (0x06)

	Command	Data[0]	Bytes Returned	Delay, ms
25-02451, 25-02452	0x06	0x00	2	1
01-02453	0x06	0x00	4	2

The revision version number of the firmware will be returned on this command.

Bit	Data[1]								Data[0]							
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Value	Board Firmware Revision Number															

Table 11-5 Returned Firmware Revision Number

Manual Reset (0x80)

Command	Data[0]	Bytes Returned	Delay, ms
0x80	0x00	0	-

If required, the user can reset the TTC node using this command. When issued, the board will reset within 1 second. This command will result in the board being brought up in its defined initial condition.

Resetting the board in this fashion will increment the Manual Reset Counter. More details about this counter are found in section 11.5.

11.4 Telemetry

The node telemetries allow the satellite's on board computer (OBC) to monitor the operation of the EPS.

Each available telemetry is represented by a two-byte code. These codes consist of:

- What type of telemetry is requested, i.e. PDM or PCM, analogue inputs, or some other form of sensor.
- The channel being requested.
- The reading to take: voltage, current, temperature etc.

A break-down of the telemetry structure and commands is given below. If a telemetry is requested which is not available, a Channel Error will be generated.

Get Telemetry (0x10)

	Command	Data[1]	Data[0]	Bytes Returned	Delay, ms
Table 11-11 Telemetries	0x10	0xE?	0x??	2	15
All other telemetries	0x10	0xE?	0x??	2	5

As described above, requesting telemetry involves sending the command 0x10 plus a 2 byte telemetry code to the node. Once transmitted, the node will configure itself to read the requested value. The general format for telemetry codes is shown in Table 11-6, and an exhaustive list of commands in Table 11-8 through Table 11-11 – refer to the table annotations for applicability to particular products. The data returned will be in the format shown in Table 11-7.

Data[1]				Data[0]			
Nibble 3		Nibble 2		Nibble 1		Nibble 0	
Family	Code	TLM Type	Code	Channel	Code	Attribute	Code
EPS	E	BCR	1	Channel Number N	N	Voltage	0
						Current A	4
						Current B	5
						Temperature A	8
						Temperature B	9
EPS	E	Main Power	2	Core Bus Miscellaneous	0 to 7 8 to F	Voltage	0
						Current	A
EPS	E	Temperature	3	Motherboard	0 to 7	Temperature	8
EPS	E	PDM	4 to 7	Switch Number N	N mod 16	Voltage	0
						Current	4

Table 11-6 Breakdown of Clyde Space telemetry code structure.

Bit	Data[1]							Data[0]								
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Value	0	0	0	0	0	0	ADC Result									

Table 11-7 ADC result return format

The result should then be converted to physical units via the conversion equations in Table 11-8. The equations provided in Table 11-8 are the theoretical equations for the system. If more accurate telemetry results are required, tailored equations are available from the test report for the individual product which will be supplied with the hardware. The advantage of using tailored equations is that they compensate for component tolerances and parasitic losses in an individual build of an EPS, however the tailored equations will vary slightly for every EPS manufactured and therefore may be different between flight and engineering model hardware.

Name	TLE Code	Description	Uncalibrated Conversion Equation	Units
IIDIODE_OUT	0xE284	BCR output current	$0.014662757 \times \text{ADC Count}$	A
VIDIODE_OUT	0xE280	BCR output voltage	$0.008993157 \times \text{ADC Count}$	V
I3V3_DRW	0xE205	3V3 Current Draw of EPS	$0.001327547 \times \text{ADC Count}$	A
ISV_DRW	0xE215	5V Current Draw of EPS	$0.001327547 \times \text{ADC Count}$	A
IPCM12V	0xE234	Output Current of 12V Bus	$0.002066632361 \times \text{ADC Count}$	A
VPCM12V	0xE230	Output Voltage of 12V Bus	$0.01349 \times \text{ADC Count}$	V
IPCMBATV	0xE224	Output Current of Battery Bus	$0.00681988679 \times \text{ADC Count}$	A
VPCMBATV	0xE220	Output Voltage of Battery Bus	$0.008978 \times \text{ADC Count}$	V
IPCM5V	0xE214	Output Current of 5V Bus	$0.00681988679 \times \text{ADC Count}$	A
VPCM5V	0xE210	Output Voltage of 5V Bus	$0.005865 \times \text{ADC Count}$	V
IPCM3V3	0xE204	Output Current of 3.3V Bus	$0.00681988679 \times \text{ADC Count}$	A
VPCM3V3	0xE200	Output Voltage of 3.3V Bus	$0.004311 \times \text{ADC Count}$	V
VSW1	0xE410	Output Voltage Switch 1	$0.01349 \times \text{ADC Count}$	V
ISW1	0xE414	Output Current Switch 1	$0.001328 \times \text{ADC Count}$	A
VSW2	0xE420	Output Voltage Switch 2	$0.01349 \times \text{ADC Count}$	V
ISW2	0xE424	Output Current Switch 2	$0.001328 \times \text{ADC Count}$	A
VSW3	0xE430	Output Voltage Switch 3	$0.008993 \times \text{ADC Count}$	V
ISW3	0xE434	Output Current Switch 3	$0.006239 \times \text{ADC Count}$	A
VSW4	0xE440	Output Voltage Switch 4	$0.008993 \times \text{ADC Count}$	V
ISW4	0xE444	Output Current Switch 4	$0.006239 \times \text{ADC Count}$	A
VSW5	0xE450	Output Voltage Switch 5	$0.005865 \times \text{ADC Count}$	V
ISW5	0xE454	Output Current Switch 5	$0.001328 \times \text{ADC Count}$	A
VSW6	0xE460	Output Voltage Switch 6	$0.005865 \times \text{ADC Count}$	V
ISW6	0xE464	Output Current Switch 6	$0.001328 \times \text{ADC Count}$	A
VSW7	0xE470	Output Voltage Switch 7	$0.005865 \times \text{ADC Count}$	V
ISW7	0xE474	Output Current Switch 7	$0.001328 \times \text{ADC Count}$	A
VSW8	0xE480	Output Voltage Switch 8	$0.004311 \times \text{ADC Count}$	V
ISW8	0xE484	Output Current Switch 8	$0.001328 \times \text{ADC Count}$	A
VSW9	0xE490	Output Voltage Switch 9	$0.004311 \times \text{ADC Count}$	V
ISW9	0xE494	Output Current Switch 9	$0.001328 \times \text{ADC Count}$	A
VSW10	0xE4A0	Output Voltage Switch 10	$0.004311 \times \text{ADC Count}$	V
ISW10	0xE4A4	Output Current Switch 10	$0.001328 \times \text{ADC Count}$	A
TBRD	0xE308	Motherboard Temperature	$(0.372434 \times \text{ADC Count}) - 273.15$	°C

Table 11-8 List of Telemetry Codes Common to All Products

Name	TLE Code	Description	Uncalibrated Conversion Equation	Units
VBCR1	0xE110	Voltage feeding BCR1	$0.009971 \times \text{ADC Count}$	V
IBCR1A	0xE114	Current BCR1, Connector SA1A	$0.0009775 \times \text{ADC Count}$	A
IBCR1B	0xE115	Current BCR1, Connector SA1B	$0.0009775 \times \text{ADC Count}$	A
TBCR1A	0xE118	Array Temp, Connector SA1A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR1B	0xE119	Array Temp, Connector SA1B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR1A	0xE11C	Sun Detector, Connector SA1A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR1B	0xE11D	Sun Detector, Connector SA1B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR2	0xE120	Voltage feeding BCR2	$0.009971 \times \text{ADC Count}$	V
IBCR2A	0xE124	Current BCR2, Connector SA2A	$0.0009775 \times \text{ADC Count}$	A
IBCR2B	0xE125	Current BCR2, Connector SA2B	$0.0009775 \times \text{ADC Count}$	A
TBCR2A	0xE128	Array Temp, Connector SA2A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR2B	0xE129	Array Temp, Connector SA2B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR2A	0xE12C	Sun Detector, Connector SA2A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR2B	0xE12D	Sun Detector, Connector SA2B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR3 ³	0xE130	Voltage feeding BCR3	$0.009971 \times \text{ADC Count}$	V
IBCR3A ⁴	0xE134	Current BCR3, Connector SA3A	$0.0009775 \times \text{ADC Count}$	A
IBCR3B	0xE135	Current BCR3, Connector SA3B	$0.0009775 \times \text{ADC Count}$	A
TBCR3A	0xE138	Array Temp, Connector SA3A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR3B	0xE139	Array Temp, Connector SA3B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR3A	0xE13C	Sun Detector, Connector SA3A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR3B	0xE13D	Sun Detector, Connector SA3B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR4	0xE140	Voltage feeding BCR4	$0.009971 \times \text{ADC Count}$	V
IBCR4A	0xE144	Current BCR1, Connector SA4A	$0.0009775 \times \text{ADC Count}$	A
IBCR4B	0xE145	Current BCR1, Connector SA4B	$0.0009775 \times \text{ADC Count}$	A
TBCR4A	0xE148	Array Temp, Connector SA4A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR4B	0xE149	Array Temp, Connector SA4B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR4A	0xE14C	Sun Detector, Connector SA4A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR4B	0xE14D	Sun Detector, Connector SA4B	$1.59725 \times \text{ADC Count}$	W/m ²

Table 11-9 List of Telemetry Codes Unique to 25-02451

³ Telemetry VBCR3 can be used to monitor the input voltage from 5V USB CHG⁴ Telemetry IBCR3A can be used to monitor the current draw from 5V USB CHG

Name	TLE Code	Description	Uncalibrated Conversion Equation	Units
VBCR1	0xE110	Voltage feeding BCR1	$0.0322581 \times \text{ADC Count}$	V
IBCR1A	0xE114	Current BCR1, Connector SA1A	$0.0009775 \times \text{ADC Count}$	A
IBCR1B	0xE115	Current BCR1, Connector SA1B	$0.0009775 \times \text{ADC Count}$	A
TBCR1A	0xE118	Array Temp, Connector SA1A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR1B	0xE119	Array Temp, Connector SA1B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR1A	0xE11C	Sun Detector, Connector SA1A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR1B	0xE11D	Sun Detector, Connector SA1B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR2	0xE120	Voltage feeding BCR2	$0.0322581 \times \text{ADC Count}$	V
IBCR2A	0xE124	Current BCR2, Connector SA2A	$0.0009775 \times \text{ADC Count}$	A
IBCR2B	0xE125	Current BCR2, Connector SA2B	$0.0009775 \times \text{ADC Count}$	A
TBCR2A	0xE128	Array Temp, Connector SA2A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR2B	0xE129	Array Temp, Connector SA2B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR2A	0xE12C	Sun Detector, Connector SA2A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR2B	0xE12D	Sun Detector, Connector SA2B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR3 ⁵	0xE130	Voltage feeding BCR3	$0.0099706 \times \text{ADC Count}$	V
IBCR3A ⁶	0xE134	Current BCR3, Connector SA3A	$0.0009775 \times \text{ADC Count}$	A
IBCR3	0xE135	Current BCR3, Connector SA3B	$0.0009775 \times \text{ADC Count}$	A
TBCR3	0xE138	Array Temp, Connector SA3A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR3B	0xE139	Array Temp, Connector SA3B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR3A	0xE13C	Sun Detector, Connector SA3A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR3B	0xE13D	Sun Detector, Connector SA3B	$1.59725 \times \text{ADC Count}$	W/m ²

Table 11-10 List of Telemetry Codes Common to 25-02452 and 01-02453

⁵ Telemetry VBCR3 can be used to monitor the input voltage from 5V USB CHG⁶ Telemetry IBCR3A can be used to monitor the current draw from 5V USB CHG

Name	TLE Code	Description	Uncalibrated Conversion Equation	Units
VBCR4	0xE140	Voltage feeding BCR4	$0.0322581 \times \text{ADC Count}$	V
IBCR4A	0xE144	Current BCR4, Connector SA4A	$0.0009775 \times \text{ADC Count}$	A
IBCR4B	0xE145	Current BCR4, Connector SA4B	$0.0009775 \times \text{ADC Count}$	A
TBCR4A	0xE148	Array Temp, Connector SA4A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR4B	0xE149	Array Temp, Connector SA4B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR4A	0xE14C	Sun Detector, Connector SA4A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR4B	0xE14D	Sun Detector, Connector SA4B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR5	0xE150	Voltage feeding BCR5	$0.0322581 \times \text{ADC Count}$	V
IBCR5A	0xE154	Current BCR5, Connector SA5A	$0.0009775 \times \text{ADC Count}$	A
IBCR5B	0xE155	Current BCR5, Connector SA5B	$0.0009775 \times \text{ADC Count}$	A
TBCR5A	0xE158	Array Temp, Connector SA5A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR5B	0xE159	Array Temp, Connector SA5B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR5A	0xE15C	Sun Detector, Connector SA5A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR5B	0xE15D	Sun Detector, Connector SA5B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR6	0xE160	Voltage feeding BCR6	$0.0322581 \times \text{ADC Count}$	V
IBCR6A	0xE164	Current BCR6, Connector SA6A	$0.0009775 \times \text{ADC Count}$	A
IBCR6B	0xE165	Current BCR6, Connector SA6B	$0.0009775 \times \text{ADC Count}$	A
TBCR6A	0xE168	Array Temp, Connector SA6A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR6B	0xE169	Array Temp, Connector SA6B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR6A	0xE16C	Sun Detector, Connector SA6A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR6B	0xE16D	Sun Detector, Connector SA6B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR7	0xE170	Voltage feeding BCR7	$0.0322581 \times \text{ADC Count}$	V
IBCR7A	0xE174	Current BCR7, Connector SA7A	$0.0009775 \times \text{ADC Count}$	A
IBCR7B	0xE175	Current BCR7, Connector SA7B	$0.0009775 \times \text{ADC Count}$	A
TBCR7A	0xE178	Array Temp, Connector SA7A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR7B	0xE179	Array Temp, Connector SA7B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR7A	0xE17C	Sun Detector, Connector SA7A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR7B	0xE17D	Sun Detector, Connector SA7B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR8	0xE180	Voltage feeding BCR8	$0.0322581 \times \text{ADC Count}$	V
IBCR8A	0xE184	Current BCR8, Connector SA8A	$0.0009775 \times \text{ADC Count}$	A
IBCR8B	0xE185	Current BCR8, Connector SA8B	$0.0009775 \times \text{ADC Count}$	A
TBCR8A	0xE188	Array Temp, Connector SA8A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR8B	0xE189	Array Temp, Connector SA8B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR8A	0xE18C	Sun Detector, Connector SA8A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR8B	0xE18D	Sun Detector, Connector SA8B	$1.59725 \times \text{ADC Count}$	W/m ²
VBCR9	0xE190	Voltage feeding BCR9	$0.0322581 \times \text{ADC Count}$	V
IBCR9A	0xE194	Current BCR9, Connector SA9A	$0.0009775 \times \text{ADC Count}$	A
IBCR9B	0xE195	Current BCR9, Connector SA9B	$0.0009775 \times \text{ADC Count}$	A
TBCR9A	0xE198	Array Temp, Connector SA9A	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
TBCR9B	0xE199	Array Temp, Connector SA9B	$(0.4963 \times \text{ADC Count}) - 273.15$	°C
SDBCR9A	0xE19C	Sun Detector, Connector SA9A	$1.59725 \times \text{ADC Count}$	W/m ²
SDBCR9B	0xE19D	Sun Detector, Connector SA9B	$1.59725 \times \text{ADC Count}$	W/m ²
TLM_TBRD_DB	0xE388	Daughterboard Temperature	$(0.372434 \times \text{ADC Count}) - 273.15$	°C

Table 11-11 List of Telemetry Codes Unique to 01-02453

11.5 Watchdogs and Reset Counters

Two on-board watchdog timers are used to restart the device if it becomes non-operational due to an error in the microcontroller. The *Communications Watchdog* is used to reset the device if a designated period passes during which the device receives no data on the I²C bus. The second watchdog is the on-board *Software Watchdog* which is used to reset the device if the microcontroller has malfunctioned. If the node determines that an error has occurred, the device is rebooted into its pre-defined initial state.

Both watchdogs have associated counters which can be queried to determine the number of times the device has reset itself through either a lack of communications or a software error. A third counter is also available which maintains a record of how many times the device is reset from a Brown-Out condition.

Get Communications Watchdog Period (0x20)

Command	Data[0]	Bytes Returned	Delay, ms
0x20	0x00	2	1

This command provides the user with the current communications watchdog timeout that has been set. The returned value is indicated in minutes.

Set Communications Watchdog Period (0x21)

Command	Data[0]	Bytes Returned	Delay, ms
0x21	<i>Period</i>	0	-

The Communications Watchdog by default has a value of 4 minutes set as its timeout period. If 4 minutes pass without a command being received, then the device will reboot into its pre-defined initial state. This value of 4 minutes can be changed using the Set Communications Watchdog Period command, **0x21**. The data byte specifies the number of minutes the communications watchdog will wait before timing out.

A minimum value of 1 minute or a maximum of 90 minutes can be set. The device will always reboot with a timeout value of 4 minutes set. If an invalid value is specified, the device will generate a Data Error.

Reset Communications Watchdog (0x22)

Command	Data[0]	Bytes Returned	Delay, ms
0x22	0x00	0	1

Any valid command will reset the communications watchdog timer. If the user does not require any telemetry from the board, this command can be sent to reset the communications watchdog.

Get Number of Brown-out Resets (0x31)

	Command	Data[0]	Bytes Returned	Delay, ms
25-02451, 25-02452	0x31	0x00	2	1
01-02453	0x31	0x00	4	2

This counter is designed to keep track of the number of brown-out resets that have occurred. This counter will roll over at 255 to 0. The first two bytes outputted represent the Motherboard's value, the second two (01-02453 only) represent the Daughterboard's.

Get Number of Automatic Software Resets (0x32)

	Command	Data[0]	Bytes Returned	Delay, ms
25-02451, 25-02452	0x32	0x00	2	1
01-02453	0x32	0x00	4	2

If the on-board microcontroller has experienced a malfunction, such as being stuck in a loop, it will reset itself into a pre-defined initial state. Using this command, **0x32**, it is possible to retrieve the number of times this reset has occurred. The first two bytes outputted represent the Motherboard's value, the second two (01-02453 only) represent the Daughterboard's. This counter will roll over at 255 to 0.

Get Number of Manual Resets (0x33)

	Command	Data[0]	Bytes Returned	Delay, ms
25-02451, 25-02452	0x33	0x00	2	1
01-02453	0x33	0x00	4	2

A count is kept of the number of times the device has been manually reset using the Reset command. Sending the command **0x33** with data byte **0x00** will return the number of times the device has been reset in this fashion. The first two bytes outputted represent the Motherboard's value, the second two (01-02453 only) represent the Daughterboard's. This counter will roll over at 255 to 0.

Get Number of Communications Watchdog Resets

Command	Data[0]	Bytes Returned	Delay, ms
0x34	0x00	2	1

As described previously, the device will reset itself if it does not receive any data via I²C for a pre-defined length of time. The communications node keeps a count of the number of times such an event has taken place. Sending the command **0x34** along with the data byte **0x00** will return the number of communication watchdog resets. This counter will roll over at 255 to 0.

11.6 PDM Control

On-board power distribution modules can either be controlled via commands which address them individually or all at once. Associated with each module is its expected status, its actual status and the state it will be initialised with when the EPS is powered on or reset.

Switch On All PDMs (0x40)

Command	Data[0]	Bytes Returned	Delay, ms
0x40	0x00	0	-

When this command is issued, all PDMs switch on.

Switch Off All PDMs (0x41)

Command	Data[0]	Bytes Returned	Delay, ms
0x41	0x00	0	-

When this command is issued, all PDMs switch off.

Get Actual State of All PDMs (0x42)

Command	Data[0]	Bytes Returned	Delay, ms
0x42	0x00	4	20

The PDMs have over-current protection built in. As a result, a PDM that is expected to be on may have tripped. This command returns the actual state of all the PDMs. The bits within the bytes returned represent the state of each PDM, with 0 representing off and 1 representing on. The order of bits is shown in Table 11-12.

Bit	7	6	5	4	3	2	1	0
Data[3]	-		-	-	-	-	-	-
Data[2]	-	-	-	-	-	-	-	-
Data[1]	-	-	-	-	-	PDM 10	PDM 9	PDM 8
Data[0]	PDM 7	PDM 6	PDM 5	PDM 4	PDM 3	PDM 2	PDM 1	-

Table 11-12 PDM Byte Codes.

Get Expected State of All PDMs (0x43)

Command	Data[0]	Bytes Returned	Delay, ms
0x43	0x00	4	1

This command returns the expected state of all the PDMs – that is, whether they have been commanded on or off, regardless of whether overcurrent protection has tripped. The format of the returned data is given by Table 11-12, with 0 representing OFF and 1 representing ON.

Get Initial State of All PDMs (0x44)

Command	Data[0]	Bytes Returned	Delay, ms
0x44	0x00	4	20

The initial state of the PDMs is returned using this command. The initial state for all the PDMs is returned in response to this command. The bit indication is the same as that in Table 11-12, with a 1 indicating the PDM is selected to be ON at power up or reset

Set All PDMs to Initial State

Command	Data[0]	Bytes Returned	Delay, ms
0x45	0x00	0	20

This command sets the initial state of the PDMs after power on or reset. This includes resetting all timers associated with each PDM (See Section 11.7 for more information about PDM Timers).

Switch PDM-N "On" (0x50)

Command	Data[0]	Bytes Returned	Delay, ms
0x50	Channel #	0	-

This command turns on an individual PDM defined in the data byte, PDM 1 is 0x01, PDM2 is 0x02 etc. If an invalid channel is specified, 0xFFFF is returned and the device will generate an *Invalid Channel* error.

Switch PDM-N "Off" (0x51)

Command	Data[0]	Bytes Returned	Delay, ms
0x51	Channel #	0	-

This command turns off an individual PDM defined in the data byte, PDM 1 is 0x01, PDM2 is 0x02 etc. If an invalid channel is specified, 0xFFFF will be returned and the device will generate an *Invalid Channel* error.

Set PDM-N's Initial State to "On" (0x52)

Command	Data[0]	Bytes Returned	Delay, ms
0x52	Channel #	0	200

Using the command **0x52** allows a PDM's initial status to be set to ON. After a reset or reboot, this PDM channel will be enabled. The channel is specified in the data byte, PDM 1 is 0x01, PDM2 is 0x02 etc. If an invalid channel is specified, 0xFFFF will be returned and the device will generate an *Invalid Channel* error.

Set PDM-N's Initial State to "Off" (0x53)

Command	Data[0]	Bytes Returned	Delay, ms
0x53	Channel #	0	200

Using command **0x53** allows a PDM's initial status to be set to OFF. After a reset or reboot, this PDM channel will be disabled. The channel is specified in the data byte, PDM 1 is 0x01, PDM2 is 0x02 etc. If an invalid channel is specified, 0xFFFF will be returned and the device will generate an *Invalid Channel* error.

Get PDM-N's Actual Status (0x54)

Command	Data[0]	Bytes Returned	Delay, ms
0x54	Channel #	2	2

The PDMs have overcurrent protection; as a result, a PDM that is expected to be on may have tripped. This command returns the actual state of the requested PDM specified in the data byte, PDM 1 is 0x01, PDM2 is 0x02 etc. A returned value of **1** indicates the PDM is ON and a returned value of **0** indicates the PDM is OFF.

11.7 PDM Timers

Each Power Distribution Module has a user configurable timer associated which allows the maximum time the PDM can be ON for to be set. Unless there is user intervention, the PDM will switch OFF after this period. This feature can be useful for high power circuitry which could drain a craft's power supply if not monitored correctly. A minimum duration of 30 seconds can be set and a maximum of 127 minutes.

Through the PDM Timer Control each PDM can be in one of three states:

- Permanently Disabled: Any attempt to enable the PDM will fail.
- Enabled **without** timer restrictions: Once switched on, the PDM will remain enabled indefinitely.
- Enabled **with** timer restrictions: Once switched on, the PDM will only remain on for a predefined period of time.

Out of the box, each PDM is set up **without** timer restrictions. Once configured the timer settings are stored in EEPROM and remain in effect even after a reboot.

Theory of Operation

Each PDM timer has two values associated with its control:

PDM Timer Limit: This is the maximum length of time the PDM will remain on for. When set to 0xFF, the timer will remain on indefinitely when enabled. If set to 0x00 the timer will always remain off, regardless of any attempt to enable it. If a command is sent to switch on a disabled channel, the error INACTIVE CHANNEL (0x04) will be generated.

The timer limit is set in multiples of 30 Seconds; therefore, supplying a value of 0x0A will set the PDM's enabled duration to 5 minutes.

Associated Commands: Set PDM Timer Limit (0x60) and Get PDM Timer Limit (0x61)

PDM Timer Current Value: The Current Value of the timer is the length of time the timer has been enabled for. Again, its values are in multiples of 30 seconds. Returned values are always rounded down. Therefore, if the PDM has been on for 7 minutes and 20 seconds the expected Timer Current Value of 0x0E would be returned.

Associated Commands: Get PDM Current Value (0x62)

If a PDM is enabled and its timer is active, sending a **Set PDM On** command to the PDM will set its current value to zero, effectively resetting the timer count. This means that from the moment a **Set PDM On** command is received the PDM will remain active for its full Timer Limit duration.

The diagram in Figure 11-13 shows the operation of the timer as a **Set PDM On** command is received.

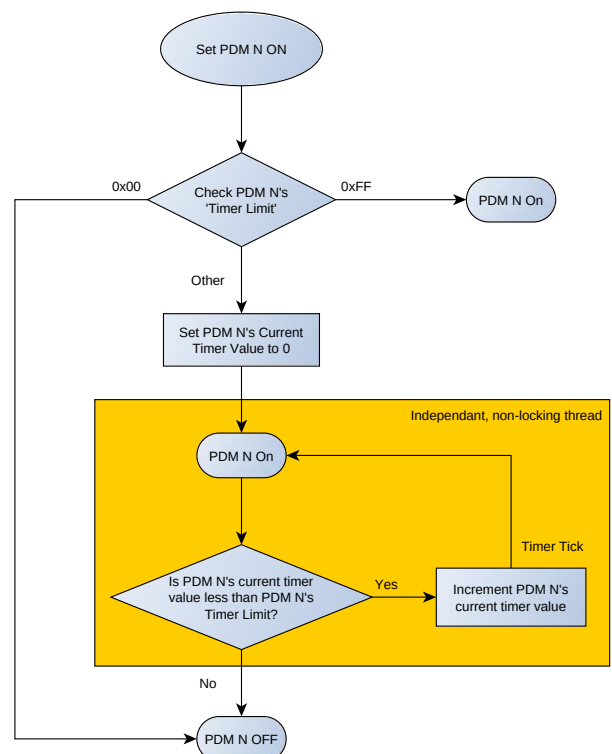


Figure 11-13 Operation of the PDM Timer State Machine

Set PDM-N's Timer Limit (0x60)

Command	Data[1]	Data[0]	Bytes Returned	Delay, ms
0x60	Channel #	Period	0	150

Set the length of time a channel can remain enabled for. The value supplied gives the duration in increments of 30 seconds, e.g. *duration=0x0A* would enable the PDM for 5 minutes.

Supplying a value of *period=0xFF* sets the PDM to remain enabled indefinitely. A value of *period=0x00* will permanently disable the PDM until such time that the timer is set to a value greater than 0x00.

If an invalid channel is specified, 0xFFFF will be returned and the device will generate an *Invalid Channel* error.

If an invalid period is specified, 0xFFFF will be returned and the device will generate an *Invalid Data* error.

Get PDM-N's Timer Limit (0x61)

Command	Data[0]	Bytes Returned	Delay, ms
0x61	Channel #	2	5

Returns the maximum timer value currently set for the PDM. Durations are returned in increments of 30 seconds, e.g. *duration=0x0A* would mean the PDM was enabled for a total of 5 minutes.

Get PDM-N's Current Timer Value (0x62)

Command	Data[0]	Bytes Returned	Delay, ms
0x62	Channel #	2	1

Returns the time passed since the PDMs timer was enabled. Durations are returned in increments of 30 seconds, e.g. *duration=0x0A* would mean the PDM has been enabled for a total of 5 minutes

11.8 PCM Control

PCM Reset (0x70)

Command	Data[0]	Bytes Returned	Delay, ms
0x70	PCM Channels	0	1

The individual power buses on the EPS can be reset using this command. Table 11-14 provides the breakdown of the data bits to reset a power bus.

Power Bus	Bit String
Battery V	0x01
5 V	0x02
3.3 V	0x04
12 V	0x08

Table 11-14 Power bus Breakdown

A combination of the bit strings can also be used. For example, to reset the 5V and the Battery V bus, send the data **0x03**.

When this command is used, the chosen power bus, or buses, will be held in reset for a period of approximately 500ms. This has the effect of turning off the power bus for this period of time.

It should be noted that when the 3.3V power bus is reset, communication to the TTC node will be lost for the period of time the bus is held in reset. The TTC node will power up in its initial configuration.

12. TEST

All EPS units are fully tested prior to shipping, and test reports are supplied. In order to verify the operation of the EPS please use the following outlined instructions.

12.1 Required Equipment

- Solar Arrays (or simulated solar array supply)
- EPS
- Battery (or simulated battery)
- Oscilloscope
- Multimeter
- Electronic Load
- Method to communicate with TTC node, eg. USB-I²C adaptor

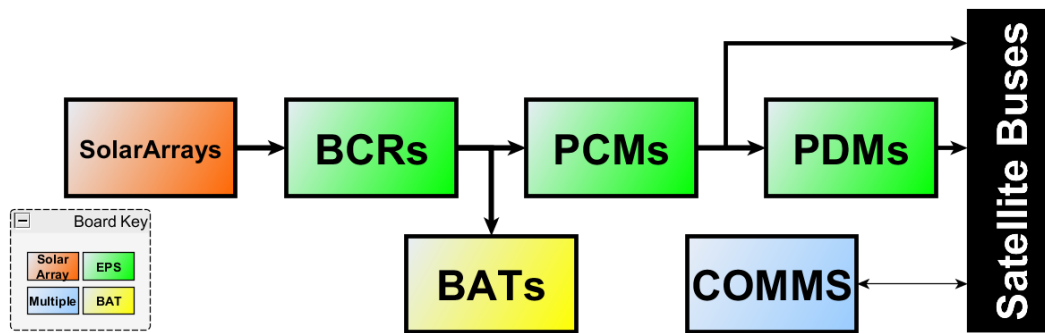


Figure 12-1 Full System Required for Test

Solar Arrays

During test phases it is not always possible to use solar arrays for testing. Other options for testing include solar array simulators or (for approximation testing) a PSU and an inline resistor.

If using a solar array simulator, it is important to ensure that the setup does not exceed the operating limits of the EPS. Table 12-1 shows the characteristics of the different compatible panel setups for the arrays.

Series Cells	V_{oc} (V)	V_{mpp} (V)	I_{sc} (mA)	I_{mpp} (mA)	Compatible BCRs
2	5.32	4.70	453.871	433.906	SEPIC BCRs
3	7.98	7.05	453.871	433.906	SEPIC BCRs
4	10.64	9.40	453.871	433.906	Buck BCRs
5	13.30	11.75	453.871	433.906	Buck BCRs
6	15.96	14.10	453.871	433.906	Buck BCRs
7	18.62	16.45	453.871	433.906	Buck BCRs
8	21.28	18.80	453.871	433.906	Buck BCRs

Table 12-1 Examples of Solar Array Configurations – Spectrolab UTJ cells @ BOL, 28°C

If a solar array simulator is not available, it is possible to approximate solar array operation with a power supply and an inline power resistor.

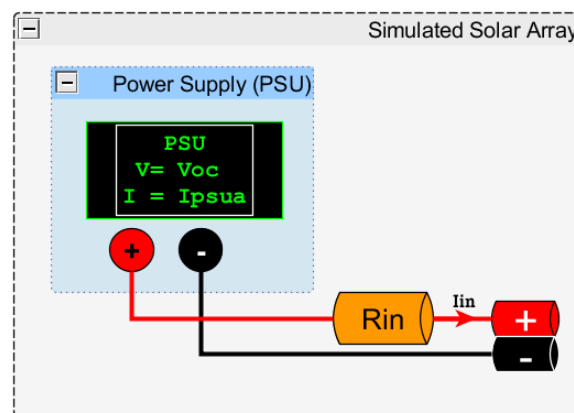


Figure 12-2 Simulated Solar Array Setup

The value of the resistor will set the current supplied and can be calculated as follows:

$$R_{in} = \frac{0.17 \times V_{oc}}{I_{in}}$$

I_{in} = the current required (normally the maximum power point current)

R_{in} = the resistance of the inline resistor selected

V_{oc} = the expected open circuit voltage of the solar array.

I_{in} is normally set, using R_{in} , to match the maximum power point current (I_{mpp}) of the expected array, but can be adjusted to simulate lower illumination conditions.

The PSU should be set using V_{oc} as the voltage setting and $2 \times I_{in}$ as the current limit (I_{psua})

Battery

During test phases it is not always possible or advisable to use a battery. For example, to test End of Charge or undervoltage shutdown operation you may want to alter the battery voltage manually rather than wait for a battery to charge/discharge. Also testing with a power supply avoids unnecessary stress on the battery from testing at high currents.

When testing without a battery, the system requires a simulated battery to be attached. This can be achieved by using a PSU (to set the battery and supply current when required/discharging) and an electronic load (to simulate the battery taking current/charging) connected in parallel.

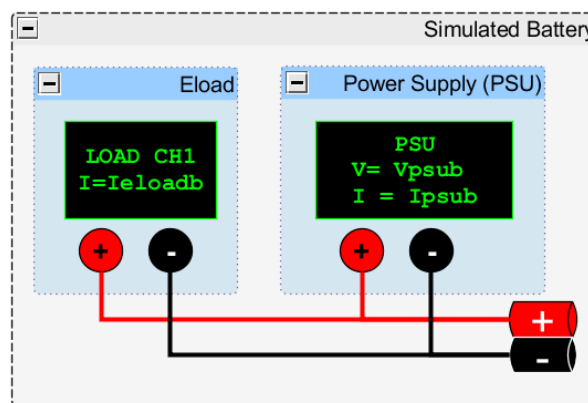


Figure 12-3 Simulated Battery Setup

The PSU should be set using the voltage as the required battery voltage (V_{psub}) and a current limit of $2C$ (I_{psub}) (the highest recommended discharge rate of the battery). The electronic load current (I_{loadb}) setting should be set to approximately $1C$ of the battery to be used. You must also ensure the eLoad setting is higher than the supplied BCR current, otherwise the BCR will be pushed into EoC.

12.2 Standalone Test Setup

The following instructions detail how to perform a test of the EPS using a simulated battery.

Before any testing commences all equipment described above should be configured with limits set up appropriately.

All PSUs should be switched off.

1. Connect the simulated battery between GND (H2.32) and PCM_IN (H2.36)
2. Place another wire between BCR_OUT (H2.42) and PCM_IN (H2.36) (make sure that the wire has enough current carrying capability).
3. Connect the solar array (or simulated solar array).

4. Switch on the simulated battery.
5. Switch on the solar array power.
6. Check that the system is operational (all power buses at expected voltages).
7. Once this has been set up it is possible to test all functions of the EPS.

For more detail on the individual tests performed on the EPS, refer to the test report which includes test setups and processes.

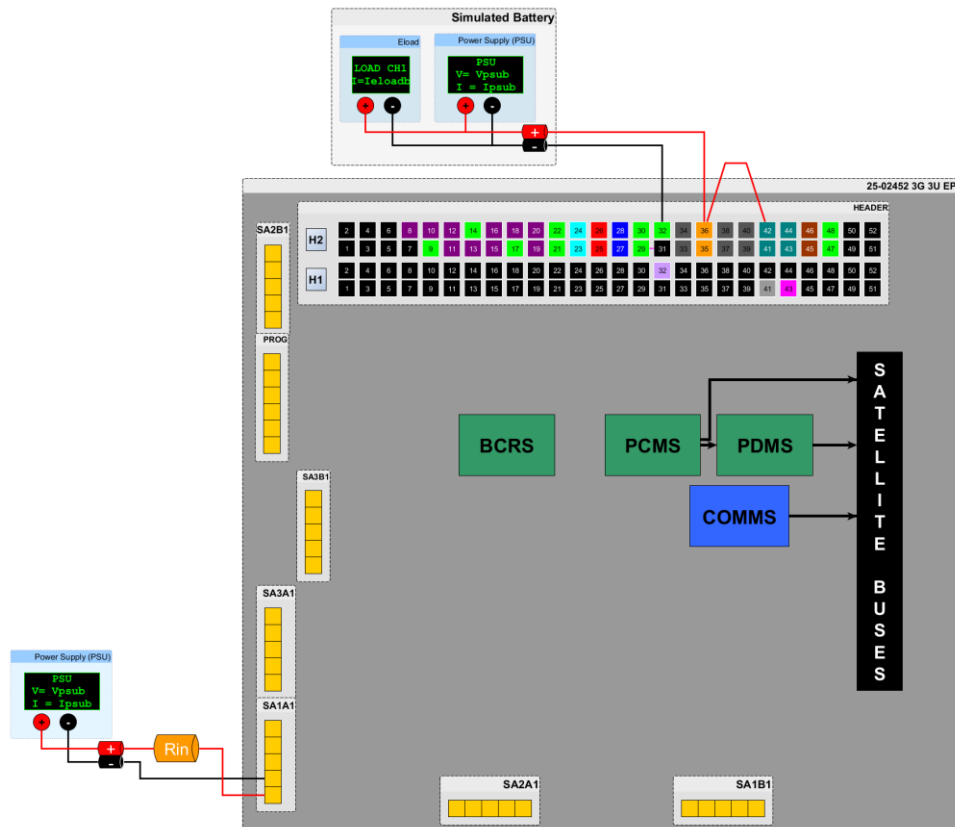


Figure 12-4 Testing a standalone EPS 25-02452. Refer to Section 8 for connection of solar array connection when testing 25-02451 or 01-02453.

12.3 Testing with Clyde Space Battery

Refer to **User manual: 3rd Generation Battery Family (USM-1192)**, Section 13 *Test* for testing with a Manned Flight/ISS Compatible battery.

13. COMPATIBLE SYSTEMS

Compatibility		Notes
Stacking Connector	CubeSat Kit Bus	
Batteries	Clyde Space Battery Systems	As listed in section 13.1.
	Other Batteries	Please contact Clyde Space
Solar Arrays	Clyde Space 2-3 cell solar array	Connects to SEPIC BCR(s)
	Clyde Space 4-8 cell solar array	Connects to Buck BCRs (not 25-02451)
	Other array technologies	Please contact Clyde Space
Structure	Clyde Space	CubeSat 1/2/3U standard structure
	Pumpkin	
	ISIS	
	Other structures	Please contact Clyde Space

Table 13-1 Compatibilities

13.1 Compatible Batteries

Standalone Batteries

This EPS is expected to be integrated with one of the following Clyde Space battery products. It is not compatible with Clyde Space 3G No Inhibits or 2G battery ranges.

Battery	Product Code
10Wh Standalone Manned Flight	01-02683
20Wh Standalone Manned Flight	01-02684
30Wh Standalone Manned Flight	01-02685
40Wh Standalone Manned Flight	01-02686

Figure 13-1 Standalone Battery Compatibility

Integrated Batteries

The 1U EPS (25-02451) is also compatible with the following integrated batteries – the battery will be integrated onto the EPS as a daughterboard, rather than having a separate PC104 header. These are not compatible with the 3U or FlexU variants.

Battery	Product Code
10Wh Integrated Manned Flight	01-02681
20Wh Integrated Manned Flight	01-02682

Figure 13-2 Integrated Battery Compatibility